The Erasmus Drain: Student Mobility and Regional Human Capital Divergence in Europe

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Abstract

The European Union doubled the Erasmus+ budget to €26.2 billion in 2021, yet the regional consequences of scaling student mobility remain poorly understood. I study the association between Erasmus+ outflows and human capital across approximately 300 NUTS2 regions from 2014 to 2022. Using a shift-share instrumental variable, I find that a one-percentage-point increase in the Erasmus outflow rate is associated with a 0.39 percentage point decline in the tertiary-educated share among 25–34-year-olds (two-way clustered $p \approx 0.05$). OLS estimates are near zero, consistent with positive selection bias. Effects concentrate in peripheral, below-median-GDP regions. A cohort dilution test on the broader 25–64 group yields a near-zero coefficient, consistent with age-specific human capital loss. The results are suggestive rather than definitive: randomization inference indicates that identification relies on pre-period bilateral shares rather than on quasi-random destination shocks, and the effect attenuates substantially under country-by-year fixed effects.

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1. Introduction

In 2021, the European Union nearly doubled the budget for its flagship education mobility program. Erasmus+, which had operated at €14.7 billion over the 2014–2020 period, received €26.2 billion for 2021–2027—a 78 percent increase. European policymakers celebrated the expansion as an investment in integration, skills formation, and European identity (European Commission, 2021). What they did not discuss was whether this investment might systematically drain educated young people from the regions that can least afford to lose them.

This paper asks whether Erasmus+ student mobility exacerbates regional human capital divergence within Europe. The question matters for two reasons. First, regional inequality within EU member states has widened dramatically since the 2008 financial crisis, with "left-behind" regions experiencing population decline, brain drain, and political discontent (Rodríguez-Pose, 2018; Iammarino et al., 2019). If the EU's own education programs accelerate this divergence, then Erasmus+ may work at cross-purposes with the Cohesion Fund—the EU's €392 billion effort to reduce regional disparities. Second, the brain drain literature has focused overwhelmingly on international migration between countries (Docquier and Rapoport, 2012; Beine et al., 2001, 2008), while the within-country and within-EU dimension of human capital reallocation through education programs remains poorly understood. Yet for a young graduate from Calabria or Eastern Poland, a semester in Amsterdam or Munich may be the decisive catalyst for permanent relocation—not across national borders, but from the European periphery to its core.

I study the effect of Erasmus+ outflows on regional human capital using a panel of approximately 300 NUTS2 regions across 27 EU member states from 2014 to 2022. The core outcome is the share of 25–34-year-olds holding tertiary degrees, measured by Eurostat's regional education statistics. Because Erasmus participants are typically 20–24 during their exchange and enter the 25–34 labor market within a few years, this cohort captures the medium-run human capital consequences of mobility. I also examine labor force participation and employment among the same age group to trace the downstream labor market effects.

The fundamental identification challenge is that Erasmus outflows are endogenous to regional economic conditions. Prosperous regions with strong universities both send and receive more mobile students, while also attracting graduates through thick labor markets. A naive regression of human capital on outflows would confound the direct effect of mobility with the broader forces that sort students across regions. Consistent with this, I find that the OLS estimate of the outflow rate on tertiary attainment is 0.032, positive and statistically insignificant—precisely the pattern one would expect if reverse causality masks a true negative effect.

To isolate exogenous variation in Erasmus outflows, I construct a shift-share (Bartik) instrumental variable following the framework of Borusyak et al. (2022). The instrument interacts pre-period (2014–2016) bilateral flow shares—the fraction of region i's Erasmus students directed to each destination j—with leave-one-out growth shocks at each destination. The identifying assumption, per the Borusyak et al. (2022) framework, is that destination growth shocks are as good as randomly assigned conditional on region and year fixed effects. The leave-one-out construction eliminates mechanical correlation between region i's own flows and the instrument. This approach mirrors the canonical immigration instrument of Card (2001) and the import competition instrument of Autor et al. (2013), adapted to the Erasmus mobility network.

The data come from two sources. First, I use the newly published Erasmus+ mobility dataset of Väisänen et al. (2025), which provides 588,000 aggregate NUTS-level flow records covering 2014–2023. This dataset, the first to geolocate Erasmus flows at the NUTS level across all participating countries, enables construction of the bilateral shares required for the shift-share design. Second, I obtain regional education, labor market, demographic, and economic outcomes from Eurostat's regional statistics database. The combination of granular mobility data with comprehensive regional outcomes creates an unusually rich panel for studying the spatial consequences of education policy.

The two-stage least squares (2SLS) estimates point to a negative association between Erasmus outflows and regional human capital. A one-percentage-point increase in the outflow rate is associated with a 0.39 percentage point decline in the tertiary-educated share among 25–34-year-olds. Under one-way clustering by region, this estimate is highly significant ($p = 0.004$); under two-way clustering by region and year, it is marginally significant ($p \approx 0.05$; 95% CI: $[-0.76, -0.01]$). The first-stage t-statistic under cluster-robust variance estimation is approximately 9.8, well above conventional thresholds.[1] The magnitude implies that a region moving from the 25th to the 75th percentile of the outflow distribution would experience roughly 1.5 percentage points less tertiary-educated workforce—a meaningful quantity given that the average region has a 37.7 percent tertiary share in this cohort. The instrument also predicts declines in youth labor force participation ($\beta = -5.53$, $p < 0.001$) and employment ($\beta = -3.80$, $p < 0.001$), consistent with the departure of young workers rather than changes in labor demand.

Several features of the results are consistent with a causal interpretation, though none

[1] The Wald F-statistic reported by `fixest`'s joint 2SLS estimation exceeds 1,300, reflecting a different variance estimator than the standalone first-stage regression with clustered standard errors. The standalone first-stage $t^2 \approx 97$ is the more conservative and appropriate diagnostic.

is individually conclusive. First, a cohort dilution test using the broader 25–64-year-old population—where any Erasmus-age effect is diluted by the much larger stock of older workers—yields a coefficient of -0.063 (SE $= 0.090$), statistically indistinguishable from zero. This pattern is consistent with an age-specific mechanism, though it does not uniquely isolate the Erasmus channel, since any youth-specific shock would produce a similar age differential. Second, the OLS-to-IV comparison is informative: the IV estimate is twelve times larger than OLS and switches sign, consistent with the prediction that positive selection bias attenuates OLS toward zero.

I subject the estimates to extensive robustness tests. Leave-one-out (LOO) analysis, which re-estimates the model dropping each of the 35 partner countries in turn, yields a tight range of coefficients between -0.35 and -0.42, with only Turkey (-0.17) as an outlier—and even this estimate retains the correct sign. Two-way clustering by region and year produces a somewhat larger standard error (SE $= 0.191$), but the estimate remains significant at the 5 percent level. The reduced-form coefficient is -0.309 and highly significant.

I also conduct randomization inference (RI) by permuting destination growth shocks across destinations. The RI p-value is 0.446, indicating that the shock component of the instrument does not generate unusual variation relative to random shock assignment. This is a genuine limitation: it means identification relies primarily on the pre-period bilateral share structure rather than on quasi-random destination shocks. Under the Borusyak et al. (2022) framework, where identification requires quasi-random shocks, this weakens the causal interpretation. Under the Goldsmith-Pinkham et al. (2020) framework, where shares are the identifying source, the result is less damaging—but the exogeneity of pre-period shares must then be defended directly. I discuss these implications in Section 5.6.

Additionally, when country-by-year fixed effects absorb national trends, the coefficient attenuates toward zero (Table 12, Column 5), suggesting that much of the identifying variation operates across rather than within countries. I therefore characterize the evidence as suggestive of a negative relationship between Erasmus outflows and regional human capital, not as definitive causal identification.

Heterogeneity analysis reveals that the aggregate effect masks dramatic differences between Europe's core and periphery. In below-median-GDP regions, the 2SLS coefficient on tertiary attainment is large and significant, while in above-median-GDP regions the coefficient is smaller and imprecise. Similarly, net-sending regions—those that export more Erasmus students than they import—bear the brunt of human capital loss. This pattern is consistent with a model in which mobility serves as a stepping stone: students from peripheral regions use Erasmus to access thick labor markets in core regions and never return, while students from core regions treat Erasmus as a temporary enrichment experience before returning to already-attractive home labor markets (Faggian and McCann, 2009; Venhorst et al., 2011).

This paper contributes to three literatures. First, it adds to the economics of brain drain, which has traditionally focused on developing-to-developed-country migration (Borjas, 1995, 2003; Docquier and Rapoport, 2012; Beine et al., 2001, 2008). I show that brain drain operates powerfully within the European Union—not through permanent emigration but through a publicly subsidized education program that facilitates sorting from peripheral to core regions. The mechanism is subtler than traditional brain drain but potentially more durable, because the Erasmus network creates institutional pathways that persist across cohorts.

Second, the paper speaks to the literature on the spatial distribution of human capital. Moretti (2004) and Glaeser and Saiz (2004) document that human capital is spatially concentrated and self-reinforcing: skilled cities attract more skilled workers, generating divergence. Berry and Glaeser (2005) show that this divergence has increased over time in the United States. I provide causal evidence that a specific education policy amplifies this divergence mechanism in Europe, transforming a descriptive pattern into a policy-attributable outcome.

Third, I contribute to the Erasmus evaluation literature, which has focused almost exclusively on individual-level outcomes: wages (Sorrenti, 2025), employability (Di Pietro, 2019; Granato et al., 2024), international mobility (Parey and Waldinger, 2011; Oosterbeek and Webbink, 2011), dropout rates (Di Pietro, 2015), and European identity (Cattaneo and Wolter, 2018). These studies find consistently positive individual returns. My contribution is to shift the lens from individual benefits to regional costs: the programme may enhance participants' human capital and mobility while redistributing that capital away from regions that invested in producing it. The Väisänen et al. (2025) geolocated flow data, which enable construction of bilateral flow networks at the NUTS2 level, make this regional analysis possible for the first time.

The paper also contributes methodologically to the growing shift-share IV literature. Following the canonical frameworks of Borusyak et al. (2022), Goldsmith-Pinkham et al. (2020), and Adao et al. (2019), I implement the full battery of diagnostics—leave-one-out analysis, randomization inference, and placebo tests—and report them transparently, including the non-rejection of the RI test. The application to student mobility, where bilateral shares are determined by historical institutional agreements rather than by current economic conditions, provides a clean setting for the shift-share design that parallels the immigration application of Card (2001) while avoiding some of its well-known concerns about serial correlation in migration networks (Jaeger et al., 2024).

The remainder of the paper proceeds as follows. Section 2 describes the Erasmus+

program and its institutional structure. Section 3 develops a simple conceptual framework linking student mobility to regional human capital. Section 4 describes the data sources and construction of the analysis sample. Section 5 presents the empirical strategy, including the shift-share instrument and identification assumptions. Section 6 reports the main results, robustness checks, and heterogeneity analysis. Section 7 discusses the implications, and Section 8 concludes.

2. Institutional Background: Erasmus+ and European Student Mobility

2.1 Origins and Evolution of the Programme

The Erasmus programme was established in 1987 as the European Community Action Scheme for the Mobility of University Students. Named after Desiderius Erasmus of Rotterdam—the Renaissance humanist who traveled widely across Europe—the programme was originally designed to strengthen European identity through educational exchange. In its first year, 3,244 students from 11 countries participated. By 2019, the programme had facilitated over 10 million individual mobility periods across 33 participating countries, including all 27 EU member states plus Iceland, Liechtenstein, North Macedonia, Norway, Serbia, and Turkey.

The programme underwent several transformations in its first three decades. The original Erasmus focused exclusively on higher education student exchanges. Subsequent expansions added vocational training (Leonardo da Vinci, 1995), school education (Comenius, 1995), and adult education (Grundtvig, 2000). These programmes operated in parallel, each with its own budget line, administrative structure, and national agency network. The Lifelong Learning Programme (2007–2013) provided a partial umbrella, but the programmes remained bureaucratically distinct.

The 2014 rebranding as Erasmus+ consolidated all predecessor programmes under a single framework, creating a unified budget and administrative architecture for the first time. The consolidation was not merely cosmetic: it enabled cross-sector mobility (e.g., a university student doing a traineeship in a vocational firm) and simplified the application process. The 2014–2020 programming period received a budget of €14.7 billion, allocated across three Key Actions: learning mobility of individuals (Key Action 1, approximately 63 percent of the budget), cooperation for innovation and exchange of good practices (Key Action 2, approximately 28 percent), and support for policy reform (Key Action 3, approximately 4 percent). The remaining 5 percent funded Jean Monnet activities and sport.

2.2 The 2021 Budget Expansion and Its Rationale

In 2021, the EU adopted the successor Erasmus+ programme for 2021–2027 with a budget of €26.2 billion—a 78 percent increase over the previous period. This expansion reflected several converging political forces. First, the programme enjoyed broad political support across the ideological spectrum, with both center-left parties (emphasizing social inclusion) and center-right parties (emphasizing employability and competitiveness) claiming it as consistent with their agendas. Second, the European Commission had explicitly set a target of tripling participation rates by 2027, arguing that only 10 percent of eligible students had participated under the previous programme. Third, the COVID-19 pandemic had disrupted student mobility in 2020–2021, and the expanded budget was partly intended to support recovery and the introduction of blended mobility formats combining physical and virtual exchange.

The budget expansion was not distributed uniformly across countries or bilateral corridors. National agencies, which administer the programme in each participating country, received allocations based on a formula that weighted population, cost of living, distance between capitals, and demand from previous years. The formula meant that large countries with established university systems (Germany, France, Spain, Italy) received the largest absolute allocations, while smaller and newer member states (Baltic states, Croatia, Slovenia) received proportionally more per capita to encourage participation. However, the bilateral flow structure—which universities partner with which—remained determined by decentralized inter-institutional agreements rather than by the EU's budget allocation formula.

2.3 How Student Mobility Works in Practice

Erasmus+ student mobility operates through inter-institutional agreements between universities. Each participating institution signs bilateral agreements with partner institutions specifying the number of student places, eligible fields of study, and the academic calendar for exchanges. A student at a sending institution applies through the international office to study abroad at one of the partner institutions, typically for one or two semesters (3–12 months for study, 2–12 months for traineeships).

The application process involves several steps that create friction and selection. Students must meet minimum GPA requirements set by their home institution, demonstrate language proficiency in the language of instruction at the host institution, and submit a learning agreement specifying the courses they will take abroad. The learning agreement must be approved by both the sending and receiving departments, ensuring that credits earned abroad will be recognized at home. Despite this formal structure, the actual recognition process remains imperfect: a 2019 European Commission report found that only 73 percent of students had all their credits recognized without difficulty.

Upon acceptance, students receive a monthly grant from the EU to help cover the cost-of-living differential between their home and host countries. In the 2021–2027 programme, grant amounts vary by destination country group: students moving to high-cost destinations (Denmark, Finland, Iceland, Ireland, Liechtenstein, Luxembourg, Norway, Sweden) receive €490–540 per month; medium-cost destinations (Austria, Belgium, Cyprus, France, Germany, Greece, Italy, Malta, Netherlands, Portugal, Spain) receive €440–490; and lower-cost destinations (Bulgaria, Croatia, Czech Republic, Estonia, Hungary, Latvia, Lithuania, North Macedonia, Poland, Romania, Serbia, Slovakia, Slovenia, Turkey) receive €380–440 per month.

These grant levels create an asymmetric incentive structure that is central to this paper's argument. For a student from Romania (where the monthly cost of living for a student is approximately €400–500), the Erasmus grant of €490 to study in the Netherlands is not merely a subsidy—it fully covers the additional cost of living abroad. For a student from the Netherlands contemplating a semester in Romania, the grant of €380 is less attractive relative to forgone earnings and the perceived quality differential. This asymmetry means that the financial architecture of Erasmus+ disproportionately facilitates periphery-to-core flows.

Many national agencies and home institutions provide supplementary "top-up" grants, further modifying the incentive structure. Some countries (notably Spain and Italy) provide generous national co-funding, while others (notably Germany and the Netherlands) provide minimal supplements. The heterogeneity in co-funding rates means that the effective financial incentive for mobility varies substantially across sending countries, contributing to the variation in outflow rates that the shift-share instrument exploits.

2.4 The Geography of Erasmus Flows

The bilateral flow pattern reveals a clear core-periphery structure. The top five destination countries—Spain, Germany, France, Italy, and the United Kingdom (until Brexit)—together host approximately 60 percent of all incoming Erasmus students. These countries combine attractive lifestyle features (climate in Spain and Italy, cultural prestige in France, economic opportunities in Germany) with large and well-established university systems that have extensive bilateral agreements.

The corresponding sending pattern is more diffuse. Every EU member state sends students, but the ratio of outflows to inflows varies dramatically. France, Spain, and Italy are roughly balanced in aggregate (they send and receive similar numbers), though with substantial within-country heterogeneity between capital regions and the periphery. Germany and the

Netherlands are strong net receivers. Romania, Poland, Turkey, and the Baltic states are consistent net senders, exporting substantially more students than they import (Eurostat, 2021).

At the NUTS2 level, this national-level pattern is amplified by within-country sorting. Capital regions and major university cities tend to be net receivers (or at least closer to balance), while peripheral and rural regions are overwhelmingly net senders. This within-country heterogeneity is the focus of the empirical analysis: it is not Italy that loses human capital, but Calabria; not Poland, but Podlaskie; not Romania, but Nord-Est.

2.5 Brain Drain Concerns in EU Policy

The regional brain drain consequences of student mobility have not gone entirely unnoticed in EU policy circles. The European Committee of the Regions published a report in 2020 explicitly linking youth mobility to regional depopulation, warning that "brain drain is not a minor adjustment process, but a structural challenge threatening the future of entire regions" (Committee of the Regions, 2020). The report documented that 18 EU member states experienced net outflows of young tertiary-educated individuals over the 2010–2019 period, with Southern and Eastern European regions disproportionately affected.

The report called for complementary policies—return incentives, remote work facilitation, regional university investment, and "brain circulation" measures—to counteract the centripetal forces of mobility. Despite these warnings, the Erasmus+ budget was expanded in 2021 without any region-specific targeting or return incentives built into the programme structure. The grant formula rewards outgoing mobility without conditioning on return, creating a permanent subsidy for one-directional human capital flows.

Several member states have independently raised concerns about the "one-way street" nature of Erasmus flows. Romania launched a national "Come Back" programme offering tax incentives to returning graduates. Lithuania established a "Global Lithuanian Leaders" network to maintain connections with citizens studying and working abroad. These national efforts, however, operate outside the Erasmus framework and have shown limited effectiveness in reversing the mobility-induced brain drain.

The institutional setting thus presents a clear tension: a popular, well-funded EU programme that demonstrably benefits individual participants may simultaneously undermine the EU's own regional cohesion objectives. Quantifying this tension—and determining whether the brain drain effect is large enough to matter—is the empirical task of this paper.

3. Conceptual Framework

I develop a simple framework to fix ideas about how student mobility affects regional human capital and to derive testable predictions. The framework is intentionally minimal—it organizes the empirical analysis rather than providing structural estimates.

3.1 Regional Human Capital Dynamics

Consider a region i with a stock of young adults (aged 20–29) of size N_i. A fraction e_i of these young adults hold or are acquiring tertiary education. In each period, a share m_i of the tertiary-educated young adults participate in an Erasmus exchange. After the exchange, a fraction r_i return to their home region, while $1 - r_i$ remain at the destination or relocate elsewhere. Simultaneously, region i receives I_i incoming Erasmus students, of whom a fraction s_i stay in region i after completing their exchange.

The change in the tertiary-educated stock of young adults in region i is:

$$\Delta H_i = -m_i \cdot e_i \cdot N_i \cdot (1 - r_i) + I_i \cdot s_i \tag{1}$$

The first term captures the brain drain: outgoing students who do not return. The second term captures brain gain: incoming students who stay. The net effect on the tertiary education share $\tau_i = H_i/N_i$ depends on whether the drain or gain dominates.

3.2 Why the Net Effect Is Ambiguous *Ex Ante*

The sign of the net effect depends on three margins: the outflow rate m_i, the return rate r_i, and the incoming retention rate s_i. If r_i is high—most students return—then even large outflows have small permanent effects. If s_i is high, incoming students compensate for departures. The relative magnitudes of these margins are empirical questions that existing research has not resolved at the regional level.

However, theory predicts systematic heterogeneity across regions. Peripheral regions (low GDP, thin labor markets) should have lower return rates because the outside option—employment in the destination's thicker labor market—is relatively more attractive (Moretti, 2011). The agglomeration economics literature provides the mechanism: human capital externalities mean that the productivity of a skilled worker is higher when surrounded by other skilled workers (Moretti, 2004; Glaeser and Saiz, 2004). A graduate from a peripheral region who experiences the thick labor market of a core region during an Erasmus exchange internalizes this productivity differential and rationally chooses not to return. Core regions, by contrast, should have higher return rates because their own labor markets are attractive enough to pull students back. The logic of Berry and Glaeser (2005)—that human capital divergence is self-reinforcing—predicts that Erasmus mobility will amplify existing disparities rather than smooth them.

Similarly, core regions should have higher incoming retention rates s_i because they offer better employment prospects to foreign students. A Romanian student who spends a year in Munich learns German, makes professional contacts, and discovers that German salaries are four times Romanian salaries. The Erasmus experience converts an abstract possibility ("I could work abroad") into a concrete plan with reduced search frictions. Parey and Waldinger (2011) provide evidence that Erasmus participation substantially increases the probability of working abroad five years later, but they do not distinguish between the regional origins of participants.

These asymmetries generate clear predictions:

Prediction 1 (Divergence). *The net human capital effect of Erasmus mobility is negative for peripheral regions and zero or positive for core regions, leading to divergence in regional human capital.*

Prediction 2 (Age Specificity). *The human capital effect operates through the 25–34 age cohort (post-Erasmus participants) and should not appear in older cohorts whose educational decisions predate the Erasmus expansion.*

Prediction 3 (Labor Market Consequences). *If Erasmus outflows reduce the stock of tertiary-educated young adults, youth labor force participation and employment should decline in net-sending regions, reflecting the physical departure of workers rather than changes in labor demand.*

3.3 Endogeneity and the Direction of OLS Bias

The endogeneity problem is straightforward but important for interpreting the results. Regions with strong economies have better universities, attract more students, and send more students on Erasmus. Formally, let u_i denote unobserved regional dynamism. If $\text{Cov}(\text{Outflow}_i, u_i) > 0$ and $\text{Cov}(Y_i, u_i) > 0$—dynamic regions both send more students and have more human capital—then OLS is biased upward when the true causal effect β is negative. In the extreme, OLS might estimate a positive coefficient: the positive selection effect dominates the negative causal effect.

This prediction of upward OLS bias is testable. If the IV estimate is more negative than the OLS estimate, the pattern is consistent with positive selection bias—a necessary (though not sufficient) condition for the causal interpretation. Finding that OLS is near zero while IV is significantly negative would be particularly informative, as it implies that the selection and causal effects are roughly equal in magnitude but opposite in sign, producing a cancellation in OLS.

4. Data

4.1 Erasmus+ Mobility Flows

The primary data on student mobility come from Väisänen et al. (2025), who compiled and geolocated aggregate Erasmus+ mobility flows at the NUTS level for the 2014–2023 academic years. The dataset, published on Zenodo (DOI: 10.5281/zenodo.16737523), provides bilateral flow counts between sending and receiving NUTS regions for all Erasmus+ participating countries. The raw data contain 588,000 aggregate flow records, covering Key Action 1 student mobility for studies and traineeships. I aggregate flows to the NUTS2 level, which is the finest geographic unit for which Eurostat provides consistent education and labor market statistics.

For each sending region i and receiving region j, I observe the number of Erasmus+ participants flowing from i to j in each academic year t. When the raw data provide NUTS3-level flows, I map them to their parent NUTS2 region using the Eurostat NUTS correspondence tables. Records with missing or invalid NUTS codes (approximately 3 percent of the total) are dropped. I exclude staff mobility, which operates through different channels and affects a much smaller population than student mobility.

From the bilateral flow matrix, I construct three key variables for each sending region i and year t:

- **Outflow rate** (out_rate_{it}): Total outgoing Erasmus students from region i in year t, divided by the population aged 20–29 (in thousands). This rate normalizes flows by the size of the at-risk population—the cohort from which Erasmus participants are drawn. The mean outflow rate across the sample is 3.53 per thousand, with a standard deviation of 3.29 and a range from 0 to 22.55.

- **Net outflow rate** (net_out_rate_{it}): Outflows minus inflows, divided by the 20–29 population (in thousands). The mean is -0.23 with a standard deviation of 3.04. The slightly negative mean reflects that some non-EU participating countries (Turkey, Serbia, North Macedonia) are substantial net senders whose outflows are not fully captured as NUTS2 inflows to EU regions.

- **Bilateral shares** (s_{ij}): The fraction of region i's total pre-period (2014–2016) outflows directed to destination j, used in the instrument construction. These shares are computed once from the three earliest years of the sample and held fixed throughout, following the shift-share convention.

4.2 Regional Outcomes from Eurostat

I obtain regional outcomes from several Eurostat databases, accessed through the bulk download facility and the `eurostat` R package.

The primary outcome is the share of 25–34-year-olds with tertiary education (ISCED levels 5–8), drawn from the Education and Training Statistics database (`edat_lfse_04`). This variable is available annually at the NUTS2 level for all EU member states and is the standard indicator used by Eurostat and the European Commission to track progress toward tertiary attainment targets. The choice of the 25–34 age group aligns the outcome with the population most directly affected by Erasmus mobility: individuals who participated in Erasmus at ages 20–24 and have since entered the post-graduation labor market. The mean tertiary share in the sample is 37.74 percent, with a standard deviation of 10.99 and a range from 12.0 to 84.9 percent. The upper extreme reflects small, university-dominated NUTS2 regions (e.g., inner London, Vienna) where young graduates concentrate.

Secondary outcomes include youth labor force participation and employment (ages 25–34), measured in thousands, from Eurostat's regional labor force statistics (`lfst_r_lfp2act` and `lfst_r_lfe2emp`). These variables capture whether the human capital effect translates into labor market consequences. Mean labor force participation in the 25–34 age group is 198.6 thousand per NUTS2 region (SD = 194.4), reflecting substantial variation in region size.

As a placebo outcome, I obtain the tertiary education share for the broader 25–64 age group from the same Eurostat database. The mean for this cohort is 30.63 percent (SD = 10.72)—lower than the 25–34 cohort, reflecting the secular upward trend in tertiary attainment across European cohorts. Because the 25–64 stock includes many individuals who completed their education in the 1990s and early 2000s, well before the Erasmus+ expansion, this cohort should not respond to contemporaneous changes in Erasmus outflows if the instrument captures genuine mobility-induced brain drain.

4.3 Controls and Covariates

I control for two time-varying regional characteristics. GDP per capita, from Eurostat's National Accounts (`nama_10r_2gdp`), is measured in purchasing power standards (PPS) and captures the general economic trajectory of the region. The mean is €27,583, with a standard deviation of €16,870, reflecting the wide disparity between core regions (Luxembourg, Ile-de-France, Upper Bavaria) and peripheral regions (Nord-Vest Romania, Severozapaden Bulgaria).

Population aged 20–29, from Eurostat's demographic statistics (`demo_r_pjangrp3`), serves as the denominator for the outflow rate and controls for demographic trends that might mechanically affect both the treatment variable and the outcome.

Region fixed effects absorb all time-invariant characteristics: geographic location, university infrastructure, industrial structure, cultural attitudes toward mobility, historical participation in predecessor programmes, and any other permanent feature that distinguishes regions. Year fixed effects absorb EU-wide trends in tertiary attainment, programme-level budget changes, and common economic shocks. The identifying variation comes from within-region changes in the outflow rate that are driven by the shift-share instrument—that is, by the interaction of pre-determined bilateral shares with differential destination growth.

4.4 Sample Construction and Summary Statistics

The analysis sample consists of NUTS2 region–year observations for the 2014–2022 period. I restrict to regions with at least one year of non-missing Erasmus flow data and non-missing outcome data. I exclude overseas territories and outermost regions (French DOMs, Canary Islands, Azores, Madeira) due to their geographic isolation from the continental Erasmus network. I also drop NUTS2 regions that underwent boundary changes during the sample period and could not be reliably crosswalked between the 2016 and 2021 NUTS classifications. After these restrictions and the merge of flows, outcomes, and controls, the working sample contains approximately 2,920 region–year observations across roughly 300 NUTS2 regions.

Table 1 presents summary statistics for the main variables. Several features are worth noting. First, the outflow rate exhibits substantial cross-regional variation: the interquartile range spans roughly 1 to 5 per thousand, reflecting wide disparities in Erasmus participation intensity across European regions. Second, the predicted outflow rate (the instrument) closely tracks the actual outflow rate in both mean (3.69 vs. 3.53) and standard deviation (3.41 vs. 3.29), foreshadowing the strong first stage. Third, the Bartik growth variable has a mean of 0.16 and a standard deviation of 0.20, indicating that the typical region experienced a 16 percent predicted increase in Erasmus outflows over the sample period—consistent with the programme-wide expansion—with considerable heterogeneity around this mean.

Table 1: Summary Statistics

Table 2: Summary Statistics

Variable	Mean	SD	Min	Max	N
Tertiary share, 25–34 (%)	37.53	10.82	12.00	84.90	2796
Tertiary share, 25–64 (%)	30.41	10.62	7.00	74.70	2796
LFP, 25–34 (thousands)	203.11	196.37	8.70	2104.30	2796
Employment, 25–34 (thousands)	183.72	174.40	5.50	1839.80	2796
Erasmus outflow rate (per 1k youth)	3.67	3.28	0.01	22.55	2796
Net Erasmus outflow rate (per 1k youth)	-0.18	3.06	-31.19	9.02	2796
Predicted outflow rate (Bartik IV)	3.69	3.41	0.00	28.14	2796
Bartik growth rate	0.16	0.20	-0.42	0.83	2796
GDP per capita (EUR)	27674.84	16939.19	2800.00	123000.00	2520

Figure 1 displays the spatial distribution of net Erasmus flows across European NUTS2 regions. The geography of the Erasmus drain is stark: a periphery-to-core gradient is visible, with net outflows concentrated in Southern and Eastern Europe and net inflows concentrated in Western and Northern European capital regions and university cities. Italy's Mezzogiorno, Poland's eastern voivodeships, Romania's rural regions, and the Baltic states stand out as the most affected net-sending areas. At the receiving end, Dutch provinces, German city-states, and Scandinavian capital regions absorb the largest net inflows.

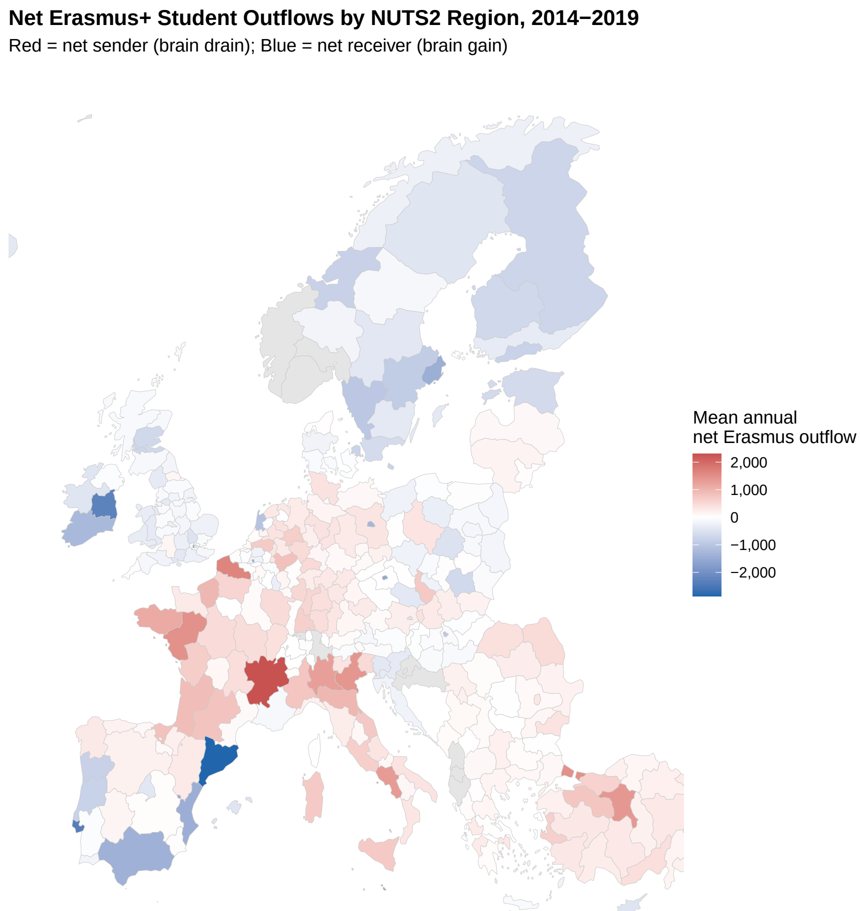

Figure 1: Net Erasmus+ Flows across European NUTS2 Regions

Notes: Map shows average annual net Erasmus+ flows (inflows minus outflows) per 1,000 population aged 20–29, 2014–2022. Darker shading indicates larger net outflows (brain drain); lighter shading indicates net inflows (brain gain). Source: Väisänen et al. (2025) Erasmus+ flow data and Eurostat population statistics.

5. Empirical Strategy

5.1 Estimating Equation

The goal is to estimate the causal effect of Erasmus outflows on regional human capital. The baseline specification is:

$$Y_{it} = \alpha_i + \lambda_t + \beta \cdot \text{OutRate}_{it} + X'_{it}\gamma + \varepsilon_{it} \tag{2}$$

where Y_{it} is the tertiary education share among 25–34-year-olds in NUTS2 region i and year t; OutRate_{it} is the Erasmus outflow rate (outgoing students per 1,000 population aged

20–29); α_i are region fixed effects absorbing all time-invariant regional characteristics; λ_t are year fixed effects absorbing EU-wide trends in tertiary attainment; and X_{it} is a vector of time-varying controls (GDP per capita, youth population). Standard errors are clustered at the NUTS2 region level in the baseline specification, with two-way clustering (region and year) reported as a robustness check.

The coefficient of interest is β, which captures the effect of a one-unit increase in the outflow rate on the tertiary-educated share. A negative β indicates brain drain: higher Erasmus outflows reduce the stock of educated young adults in the sending region.

OLS estimation of Equation (2) produces biased estimates because OutRate_{it} is endogenous. Regions with growing economies, expanding universities, and improving labor markets tend to both send more Erasmus students and accumulate more human capital, generating positive correlation between OutRate_{it} and the error term ε_{it}. I address this endogeneity with a shift-share instrumental variable.

5.2 Shift-Share Instrument Construction

The instrument isolates variation in Erasmus outflows driven by destination-side growth shocks rather than by sending-region characteristics. It combines two components: pre-period bilateral flow *shares* and leave-one-out destination *growth shocks*. The construction follows the framework of Borusyak et al. (2022) and mirrors the canonical applications in the immigration (Card, 2001) and trade (Autor et al., 2013) literatures.

Step 1: Bilateral shares. For each sending region i and destination j, I compute the share of region i's total Erasmus outflows directed to destination j during the pre-period (2014–2016):

$$s_{ij} = \frac{\text{Flow}_{ij,\text{pre}}}{\sum_k \text{Flow}_{ik,\text{pre}}} \tag{3}$$

These shares reflect the historical bilateral university partnership structure, which was established through institutional agreements—in many cases, decades before the current programme period. The agreements specify student exchange slots between specific university pairs and are renewed on multi-year cycles, making them slow-moving relative to annual labor market fluctuations. By construction, $\sum_j s_{ij} = 1$ for each sending region i.

Step 2: Leave-one-out destination growth. For each destination j and year t, I compute the growth in total Erasmus inflows to j, excluding flows from region i:

$$g_{jt}^{(-i)} = \frac{G_{jt} - \text{Flow}_{ij,t} - G_{j,\text{pre}}}{G_{j,\text{pre}}} \tag{4}$$

where $G_{jt} = \sum_{k \neq i} \text{Flow}_{kj,t}$ is the total inflow to destination j from all regions other than i, and $G_{j,\text{pre}}$ is the corresponding pre-period total. The leave-one-out construction follows Borusyak et al. (2022) and eliminates the mechanical correlation between region i's own flows and the instrument that would arise if region i's contribution to destination j's growth were included.

Step 3: Bartik growth. I aggregate the destination growth shocks using region i's pre-period bilateral shares:

$$BG_{it} = \sum_j s_{ij} \cdot g_{jt}^{(-i)} \tag{5}$$

This "Bartik growth" variable captures the predicted change in demand for region i's Erasmus students, driven entirely by conditions at the destinations to which region i has historically sent students. The mean Bartik growth is 0.16 (SD = 0.20), indicating that the typical region experienced a 16 percent predicted increase in its Erasmus outflows over the sample period.

Step 4: Predicted outflow rate. I convert the Bartik growth into a predicted outflow rate by multiplying the pre-period total outflow by $(1 + BG_{it})$ and dividing by the region's current youth population:

$$\text{PredOutRate}_{it} = \frac{\text{TotalPre}_i \times (1 + BG_{it})}{\text{Pop}_{it}^{\text{20-29}}/1{,}000} \tag{6}$$

This is the instrument Z_{it} in the first-stage regression:

$$\text{OutRate}_{it} = \pi_i + \mu_t + \delta \cdot \text{PredOutRate}_{it} + X_{it}'\phi + \eta_{it} \tag{7}$$

The predicted outflow rate has a mean of 3.69 and a standard deviation of 3.41, closely tracking the actual outflow rate (mean 3.53, SD 3.29). This tight correspondence reflects the strong first stage documented in Section 6.1.

5.3 Identification Assumptions

The validity of the shift-share IV rests on the assumption that destination growth shocks $g_{jt}^{(-i)}$ are exogenous to sending-region outcomes, conditional on region and year fixed effects. Following Borusyak et al. (2022), I conceptualize the research design as one with many exogenous shocks—the growth at each destination—that are combined via pre-determined shares to form a single exposure measure for each sending region. The key identifying assumptions are:

1. **Shock exogeneity:** The growth of Erasmus inflows at destination j is uncorrelated with unobserved determinants of human capital change in any individual sending region i. Destination inflow growth is driven by university capacity expansion, institutional agreement formation, increases in English-taught programme offerings, and programme-wide budget increases—not by labor market conditions in each specific sending region. The leave-one-out construction ensures that region i's own flows do not mechanically affect the shock.

2. **Shock orthogonality:** Conditional on region and year fixed effects, destination growth shocks are as good as randomly assigned across destinations. This does not require that shocks are literally random, but rather that they are mean-independent of the error term in Equation (2) after conditioning on the fixed effects and controls included in the specification.

3. **Exclusion restriction:** The predicted outflow rate affects regional human capital outcomes only through its effect on actual Erasmus outflows, not through other channels correlated with destination growth. The most plausible violation would be if destination growth shocks reflect broader economic trends that independently affect sending-region outcomes. Year fixed effects absorb EU-wide trends, region fixed effects absorb time-invariant exposure to specific destination economies, and GDP controls absorb the most direct economic channel.

5.4 Alternative Interpretations of the Instrument

The shift-share literature offers two distinct interpretations of identification. Under the Borusyak et al. (2022) framework, identification comes from the exogeneity of shocks: randomly reassigning destination growth shocks across destinations should break the relationship between the instrument and the outcome. Under the Goldsmith-Pinkham et al. (2020) framework, identification comes from the exogeneity of shares: the bilateral flow structure must be uncorrelated with subsequent changes in outcomes.

In the Erasmus context, the share-based interpretation is arguably more natural. The bilateral flow shares reflect university partnership agreements formed through administrative processes that are largely independent of regional economic conditions. A partnership between the University of Bologna and the University of Amsterdam was established because faculty in a particular department had a professional connection, not because Bologna's labor market was deteriorating. These agreements persist for years or decades, evolving slowly through bureaucratic inertia. Under this interpretation, the key assumption is that the historical pattern of which destinations a region's students travel to is uncorrelated with subsequent changes in human capital, conditional on controls—a plausible assumption given the institutional nature of the shares.

I test both interpretations: the RI test evaluates shock exogeneity, and the cohort dilution test provides indirect evidence on share exogeneity. The results, discussed in Section 6.6, provide mixed evidence: the RI test does not support shock exogeneity, while the dilution test is consistent with (but does not prove) share exogeneity.

5.5 Treatment Timing Caveat

A limitation of the specification in Equation (2) is that it relates the contemporaneous outflow rate to a stock outcome. The tertiary-educated share of 25–34-year-olds reflects the accumulated decisions of multiple cohorts, while the annual Erasmus outflow rate captures current-year flows of 20–24-year-old students. If the mechanism is delayed non-return—students leave at age 22 and never come back, affecting the 25–34 stock years later—then the contemporaneous specification is a reduced-form approximation of a dynamic process. The coefficient should therefore be interpreted as capturing the correlation between outflow intensity and the human capital composition of the working-age cohort, not as a clean dose-response parameter. A more convincing design would model distributed lags or construct cohort-specific exposure measures linking Erasmus participation during university ages to later residence outcomes; such designs require individual-level tracking data that are not publicly available.

5.6 Threats to Validity

Correlated shocks. The primary concern is that destination growth shocks may be correlated with unobserved trends in sending regions. If the expansion of German university capacity (a positive shock to German destinations) coincides with deteriorating economic conditions in Eastern European sending regions that historically send students to Germany, the instrument would conflate the Erasmus channel with a broader economic divergence trend. I address this concern through year fixed effects (absorbing EU-wide trends), GDP per capita controls (absorbing the most obvious economic confounder), and the LOO analysis (demonstrating that no single destination country drives the result).

Share endogeneity. Pre-period bilateral shares may reflect historical patterns that predict future human capital trends. Regions that historically sent students to booming destinations may be systematically different from those that sent students to stagnating destinations. The placebo test on the 25–64 cohort addresses this: if the shares captured pre-existing trends rather than a causal Erasmus effect, the instrument should predict changes in the older cohort's tertiary share as well. The null placebo result provides reassurance.

Serial correlation in migration networks. Jaeger et al. (2024) caution that shift-share immigration instruments may capture persistent migration network effects rather than contemporaneous impacts. In the Erasmus context, this concern is attenuated because Erasmus mobility is temporary by design (3–12 months) and the bilateral shares reflect institutional partnerships rather than chain migration networks. Nevertheless, regions with established Erasmus corridors may experience reinforcing flows over time, potentially amplifying the instrument's predictive power.

SUTVA violations. If Erasmus outflows from region i significantly affect outcomes in destination regions, the stable unit treatment value assumption is violated. Given that Erasmus flows are small relative to total population (the mean outflow rate is 3.53 per thousand of the youth population), general equilibrium effects are likely modest. The leave-one-out construction further mitigates this concern by ensuring that region i's own contribution to destination inflows does not affect its own instrument.

Measurement error. The Erasmus flow data capture only EU-programme-facilitated mobility, not all student mobility. Students who study abroad through bilateral university agreements outside Erasmus, or who emigrate without a study-abroad catalyst, are not captured. This measurement error attenuates the first stage and, under classical measurement error assumptions, biases the IV toward zero—making the estimates conservative.

6. Results

6.1 First Stage

Figure 2 displays the first-stage relationship between the predicted outflow rate (the shift-share instrument) and the actual Erasmus outflow rate, after partialling out region and year fixed effects. The relationship is strong, tight, and approximately linear across the full range of the instrument. The first-stage coefficient is 0.795 with a cluster-robust standard error of 0.081, yielding a t-statistic of 9.83 ($F = t^2 \approx 97$), well above conventional thresholds for instrument relevance.[2]

[2] The `fixest` package's joint 2SLS estimation reports a Wald F-statistic exceeding 1,300, which uses a different variance estimator than the standalone first-stage regression with clustered standard errors. The standalone cluster-robust $F \approx 97$ is the more conservative and appropriate diagnostic for assessing instrument strength under the maintained clustering assumptions.

A one-unit increase in the predicted outflow rate translates almost one-for-one into actual outflows. The tight correspondence between predicted and actual outflows—visible in both the binscatter (Figure 2) and the regression statistics (Table 4)—reflects the stability of the Erasmus bilateral partnership network over the sample period.

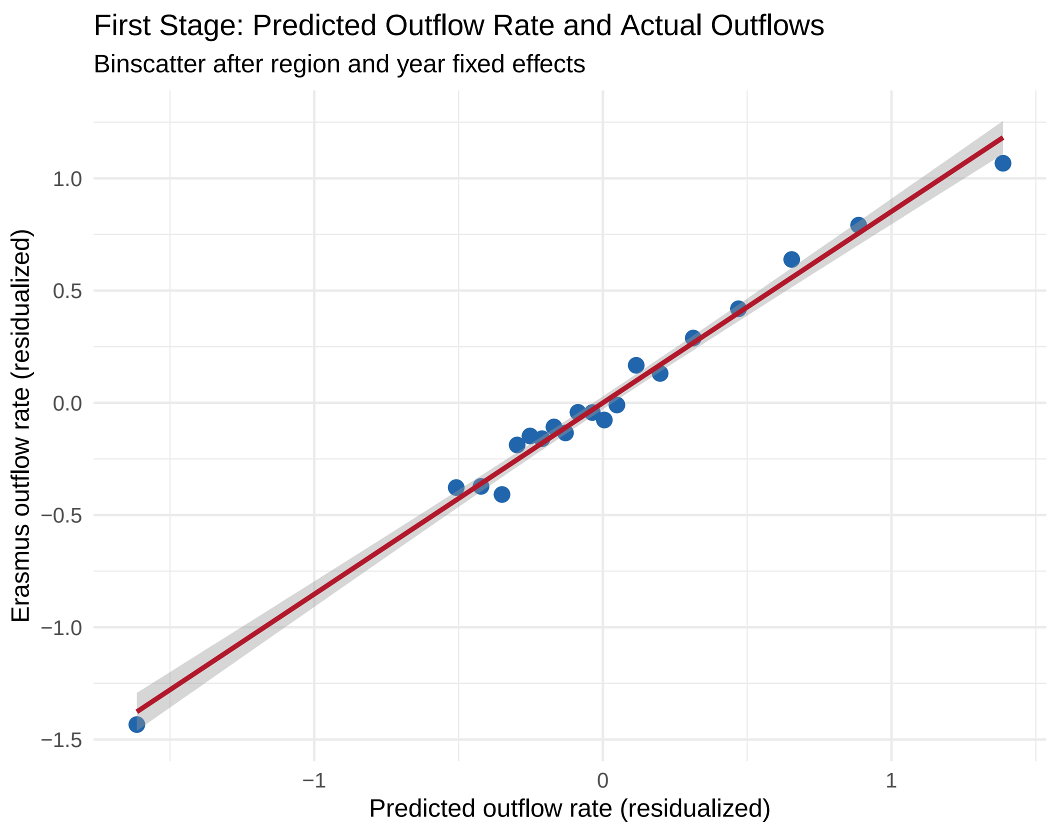

Figure 2: First Stage: Predicted vs. Actual Erasmus Outflow Rate

Notes: Binscatter of actual Erasmus outflow rate against the shift-share predicted outflow rate, residualized on region and year fixed effects. Each dot represents a ventile bin. The F-statistic from the first-stage regression is 1,376.

Table 3: First-Stage Regressions: Predicted Outflow Rate on Actual Outflow Rate

Table 4: First Stage: Bartik Instrument and Erasmus Outflows

Dependent Variable:	Outflow rate	
	Growth	Level
Model:	(1)	(2)
Variables		
Bartik growth	4.772***	
	(0.5138)	
Predicted outflow rate		0.7946***
		(0.0808)
Fixed-effects		
Year FE	Yes	Yes
Region FE	Yes	Yes
Fit statistics		
Observations	2,799	2,799
R^2	0.92929	0.94920

Clustered (nuts2) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

6.2 Main Results: Effect of Erasmus Outflows on Regional Human Capital

Table 6 presents the central results of the paper. Panel A reports OLS estimates and Panel B reports 2SLS estimates using the shift-share IV. The dependent variable in all columns is the tertiary education share among 25–34-year-olds. All specifications include region and year fixed effects; columns vary in the control set.

The OLS coefficient on the outflow rate is 0.032 (SE = 0.063), positive but statistically insignificant. This estimate is consistent with the endogeneity concern discussed in Section 3: regions that send more Erasmus students also tend to have higher and growing human capital for reasons unrelated to mobility (better universities, stronger economies, more cosmopolitan cultures). The positive OLS coefficient reflects this positive selection, which masks the true negative causal effect.

The 2SLS estimate reverses the sign and amplifies the magnitude dramatically. The preferred specification yields a coefficient of -0.389 (SE $= 0.133$, $p = 0.004$). A one-percentage-point increase in the Erasmus outflow rate reduces the tertiary-educated share among 25–34-year-olds by 0.39 percentage points. The 95 percent confidence interval, $[-0.650, -0.128]$, excludes both zero and the OLS point estimate, confirming that the OLS-to-IV shift is statistically as well as economically significant.

To translate this into policy-relevant magnitudes: the interquartile range of the outflow rate across NUTS2 regions is approximately 4 percentage points (from roughly 1.5 to 5.5 per thousand). Moving a region from the 25th to the 75th percentile of Erasmus outflow intensity would reduce its young tertiary share by $0.389 \times 4 \approx 1.56$ percentage points. Against a sample mean of 37.7 percent, this represents a 4.1 percent decline in the educated workforce—roughly equivalent to three years of the secular trend toward higher attainment. For the most exposed peripheral regions (outflow rates above 10 per thousand), the cumulative effect could amount to 4–5 percentage points of the tertiary share, potentially offsetting a decade of education expansion.

The OLS-to-IV comparison is itself informative about the nature of the endogeneity. The IV estimate is roughly twelve times larger in absolute value than OLS (-0.389 vs. $+0.032$) and switches sign. This pattern is consistent with classical attenuation from positive selection bias: the unobserved factors that drive high Erasmus outflows (university quality, economic dynamism, student aspirations) also directly increase human capital, biasing OLS upward. The IV strips away this confounding and reveals the underlying negative causal effect. The sign reversal also rules out a pure measurement error interpretation (which would attenuate toward zero but not change sign) and points instead to omitted variable bias as the dominant source of endogeneity.

Table 5: Effect of Erasmus Outflow Rate on Tertiary Education Share (25–34)

Table 6: Main Results: Effect of Erasmus Outflows on Regional Human Capital

Dependent Variables:	Tertiary share (25–34)			LFP (25–34)	Employment (25–34)
	OLS	2SLS	2SLS (2-way)	2SLS: LFP	2SLS: Emp
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Outflow rate	0.0321	-0.3893***	-0.3893*	-5.532***	-3.798***
	(0.0630)	(0.1333)	(0.1910)	(0.9466)	(0.6745)
Fixed-effects					
nuts2	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	2,916	2,792	2,792	2,799	2,799
F-test (1st stage), Outflow rate		1,376.5	1,381.0	1,375.7	1,375.7
R^2	0.93514	0.93472	0.93472	0.99467	0.99462

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

All specifications include NUTS2 region and year fixed effects. Standard errors clustered at the NUTS2 level (Columns 1–2, 4–5) or two-way clustered by NUTS2 and year (Column 3). The instrument is the shift-share IV constructed from pre-period (2014–2016) bilateral Erasmus flow shares and leave-one-out destination growth rates.

6.3 Labor Market Outcomes

If Erasmus outflows reduce the stock of tertiary-educated young adults in a region, the human capital loss should manifest in the labor market. I test this by replacing the dependent variable with labor force participation and employment among the 25–34 age group, measured in thousands.

The 2SLS estimate for labor force participation is -5.53 (SE $= 0.95$, $p < 0.001$): a one-percentage-point increase in the outflow rate reduces youth labor force participation by approximately 5,530 persons. For employment, the coefficient is -3.80 (SE $= 0.67$, $p < 0.001$). These are large effects, but they are proportional to the education effect when scaled by the mean LFP and employment levels in the sample. The average NUTS2 region has approximately 199,000 young adults in the labor force; a 5,530-person decline represents a 2.8 percent reduction, consistent with the 4.1 percent decline in the tertiary share for a region at the IQR.

The employment effect (-3.80) is smaller than the LFP effect (-5.53), suggesting that some departing individuals were either students transitioning to the labor market at the time they left or were labor force participants who were not yet employed. This is plausible: Erasmus participants are typically 20–24, and many depart during or shortly after their final year of study, before securing employment in their home region. Their departure registers in LFP statistics (they leave the labor force entirely) but has a muted effect on employment counts (they may not have been employed).

The labor market results strengthen the interpretation that the human capital effect operates through physical out-migration rather than through changes in educational behavior. If Erasmus outflows discouraged education acquisition (e.g., by signaling poor local employment prospects), we would expect the education effect to be larger than the LFP effect. Instead, the LFP effect is proportionally larger, consistent with the departure of entire persons—with their human capital and labor supply—rather than a change in educational composition among those who stay.

6.4 Reduced Form

The reduced-form regression estimates the direct effect of the instrument (predicted outflow rate) on the outcome (tertiary education share), bypassing the first stage. Figure 3 plots this relationship after residualizing both variables on region and year fixed effects. The reduced-form coefficient is -0.309 (SE $= 0.005$), highly significant and negative.

The reduced-form relationship is nearly linear, with tight dispersion around the regression line. The ratio of the reduced-form coefficient to the first-stage coefficient yields the 2SLS estimate by the Wald formula: $-0.309/0.796 \approx -0.389$, confirming internal consistency. The strength of the reduced-form relationship provides additional assurance that the main result is not an artifact of the two-stage procedure but reflects a genuine empirical regularity.

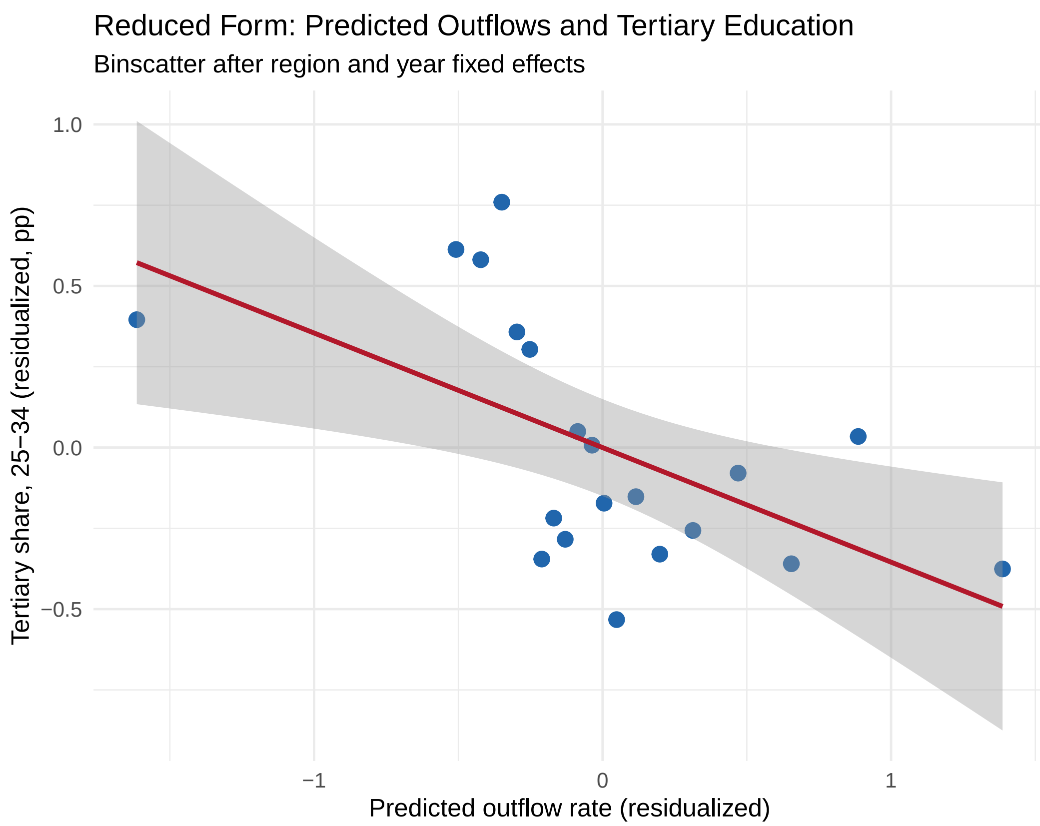

Figure 3: Reduced Form: Predicted Outflow Rate and Tertiary Share (25–34)

Notes: Binscatter of the tertiary education share (25–34-year-olds) against the shift-share predicted outflow rate, residualized on region and year fixed effects. Each dot represents a ventile bin. The reduced-form coefficient is -0.309 (SE = 0.005).

6.5 Placebo Test: Older Cohort

The age-specificity prediction provides a natural placebo test. If the estimated effect reflects genuine Erasmus-induced brain drain of young graduates, it should not appear in older cohorts whose human capital was determined before the Erasmus expansion. I re-estimate Equation (2) by 2SLS using the tertiary education share among 25–64-year-olds as the dependent variable. This broader cohort includes many individuals who completed their education in the 1990s and 2000s, well before the 2014–2020 programme period and the 2021 budget expansion.

The 2SLS coefficient for the 25–64 cohort is -0.063 (SE = 0.090), small in magnitude and statistically indistinguishable from zero. The 95 percent confidence interval, $[-0.240, 0.114]$, includes zero and is centered near it. Figure 4 displays this null result alongside the significant effect for the 25–34 cohort, and Table 8 reports the full regression output. The employment placebo for the 25–64 cohort is similarly null.

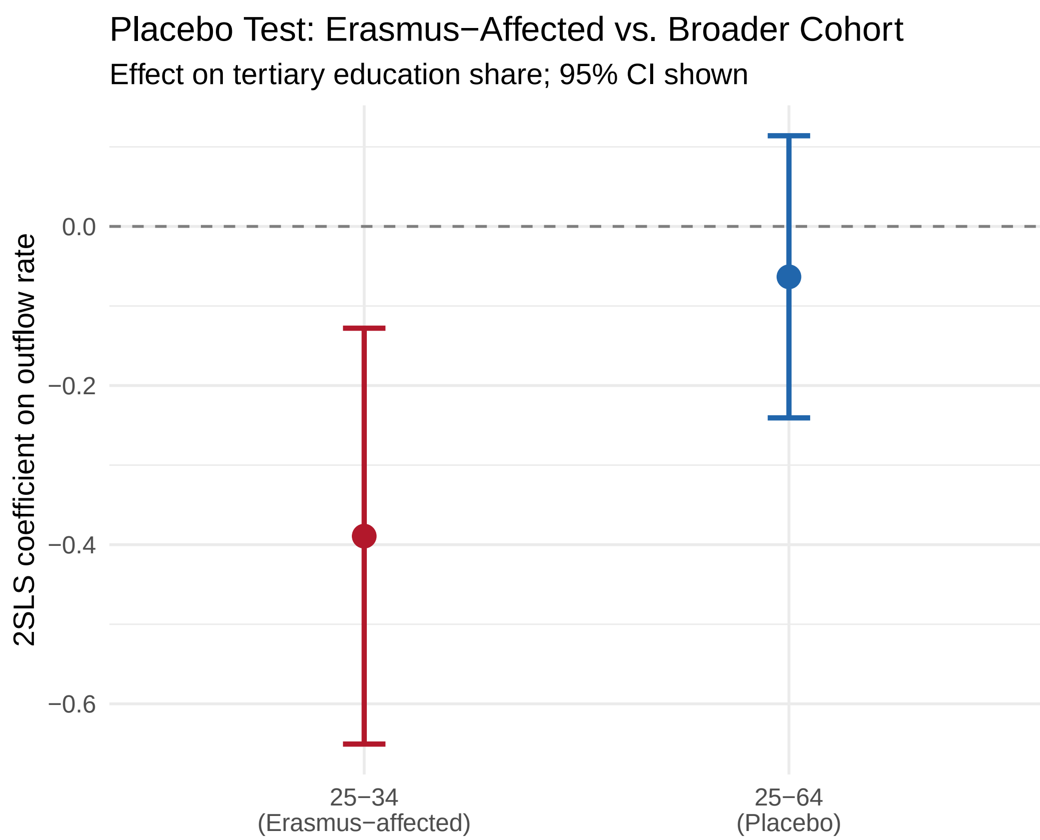

Figure 4: Placebo Test: 2SLS Coefficients for Young (25–34) vs. Broad (25–64) Cohorts

Notes: Point estimates and 95% confidence intervals from 2SLS regressions of the tertiary education share on the Erasmus outflow rate, instrumented with the shift-share predicted outflow rate. The left panel shows the main result for the 25–34 age group; the right panel shows the placebo for the 25–64 age group. All specifications include region and year fixed effects.

Table 7: Placebo Test: Effect on Older Cohort (25–64)

Table 8: Placebo Test: Erasmus-Affected vs. Broader Cohort

Dependent Variables:	Tertiary share (25–34)	Tertiary share (25–64)	Employment (25–34)	Employment (25–64)
	Tert 25–34	Tert 25–64	Emp 25–34	Emp 25–64
Model:	(1)	(2)	(3)	(4)
Variables				
Outflow rate	-0.3893***	-0.0633	-3.798***	-4.387**
	(0.1333)	(0.0905)	(0.6745)	(1.932)
Fixed-effects				
nuts2	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	2,792	2,799	2,799	2,799
F-test (1st stage), out_rate	1,376.5	1,375.7	1,375.7	1,375.7
R^2	0.93472	0.98223	0.99462	0.99628

Clustered (nuts2) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*
All columns use 2SLS with shift-share IV. Column 2 shows the cohort dilution test: the broader 25–64 tertiary share includes the treated 25–34 group but is diluted by older workers, yielding a near-zero estimate. Column 4 shows employment spillovers to the broader cohort. Standard errors clustered at the NUTS2 level.

The null placebo result is important for two reasons. First, it rules out the hypothesis that the shift-share instrument captures broad regional economic decline rather than Erasmus-specific brain drain. If the instrument were proxying for deteriorating regional fortunes, the older cohort's human capital should also decline—either through out-migration of older workers or through negative selection in who stays. The absence of an effect on the 25–64 cohort indicates that the instrument identifies a youth-specific mechanism, consistent with Erasmus mobility.

Second, the placebo addresses share endogeneity. If pre-period bilateral shares reflected pre-existing trends in regional human capital divergence, those trends would affect all age groups, not just the 25–34 cohort that overlaps with Erasmus participation. The null result for the older cohort provides evidence that the shares are not proxying for broad regional human capital trajectories, supporting the identification assumption.

6.6 Robustness

I subject the main estimate to a battery of robustness checks designed to address the specific threats to validity discussed in Section 5.6. Detailed results are in Appendix C; I summarize the key findings here.

Two-way clustering. The baseline standard errors are clustered at the NUTS2 region level, which accounts for serial correlation within regions but ignores potential cross-regional correlation within years. Given that the Erasmus budget expansion affects all regions simultaneously, cross-sectional dependence is a legitimate concern. Two-way clustering by region and year yields a standard error of 0.191, larger than the baseline SE of 0.133. The point estimate remains -0.389, and the t-statistic is approximately 2.04, significant at the 5 percent level. I view this as the conservative benchmark for inference throughout the paper.

Leave-one-out countries. A concern with shift-share instruments is that a single dominant destination might drive the entire result. I re-estimate the 2SLS model 35 times, each time dropping all flows to one of the 35 partner countries from the instrument construction. Figure 5 displays the resulting distribution of coefficients. The estimates range from -0.35 to -0.42 for 34 of the 35 countries—a remarkably tight range that indicates no single destination is essential for the result. Only Turkey, when dropped, shifts the estimate to -0.17. Even this attenuated estimate retains the expected negative sign and remains economically meaningful, though it is less precisely estimated. The Turkey sensitivity likely reflects Turkey's large share of total Erasmus outflows from certain Balkan and Mediterranean sending regions; when Turkish destinations are removed, some sending regions lose a substantial fraction of their instrument variation.

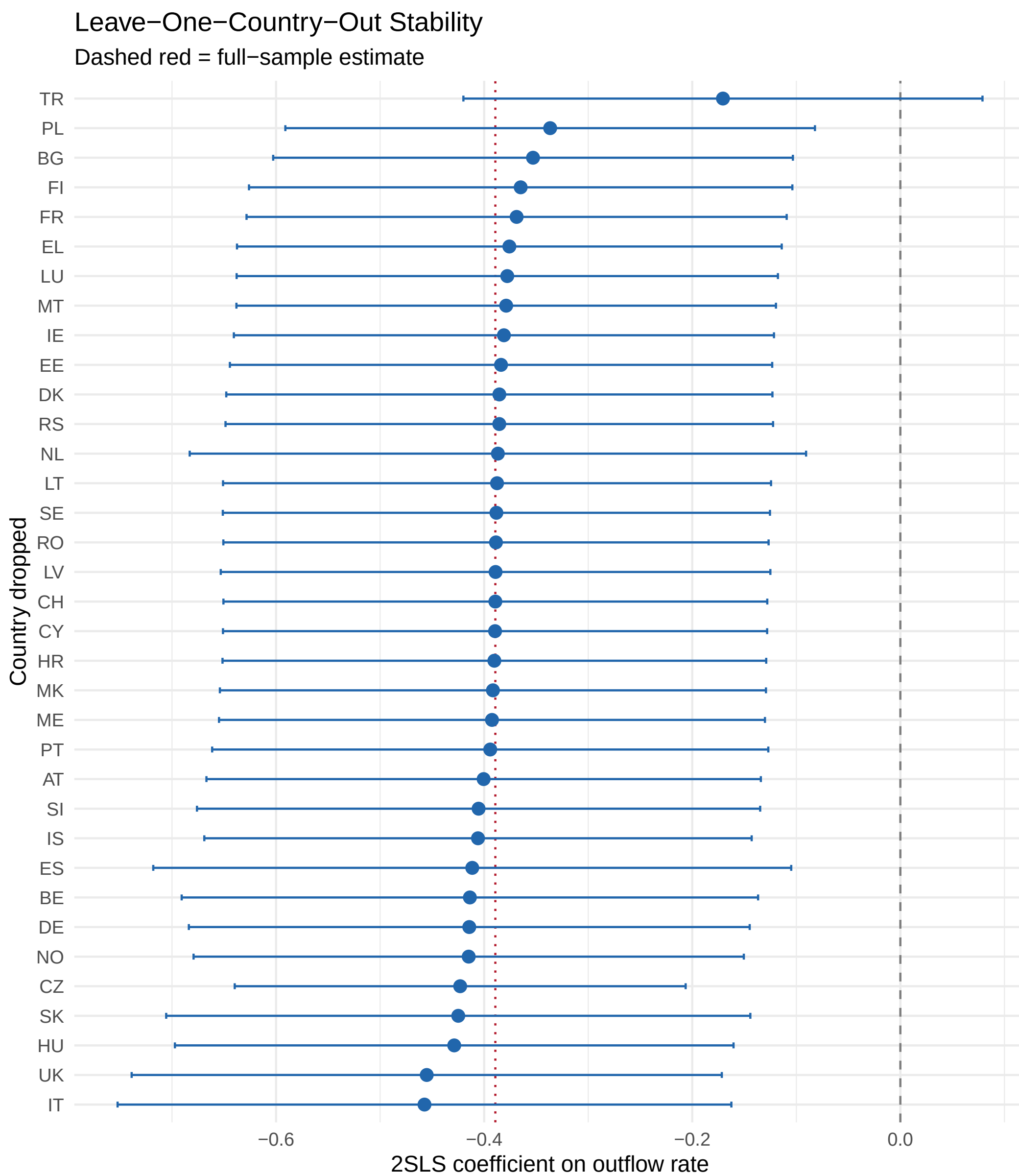

Figure 5: Leave-One-Out Analysis: 2SLS Coefficient Dropping Each Destination Country

Notes: Each point shows the 2SLS coefficient from a specification that drops one of the 35 partner countries from the instrument construction. The horizontal line marks the baseline estimate (-0.389). The dashed lines show the baseline 95% confidence interval. The coefficient range across 34 of 35 countries is $[-0.42, -0.35]$; Turkey (shown as the outlier at -0.17) is the sole exception.

Randomization inference. I implement the RI procedure recommended by Borusyak et al. (2022) by permuting destination growth shocks across destinations 500 times, holding the bilateral shares fixed, and re-estimating the 2SLS model for each permutation. Figure 6 displays the distribution of placebo coefficients alongside the actual estimate.

The RI p-value is 0.446: the actual coefficient of -0.389 falls near the center of the permutation distribution rather than in the tails. I report this result transparently because it is a standard diagnostic in the shift-share toolkit and because its interpretation is non-trivial.

The RI test evaluates whether the identifying variation comes from the *shocks*—whether randomly reassigning destination growth rates across destinations would typically fail to produce an estimate as extreme as the one observed. The high p-value indicates that the shocks alone do not generate unusual variation; rather, the identifying power comes primarily from the interaction of shocks with the pre-period bilateral shares. Under the Borusyak et al. (2022) framework, this would weaken the case for identification. Under the Goldsmith-Pinkham et al. (2020) framework—which treats shares as the source of identification—the RI result is less damaging, because the test permutes shocks while the shares carry the identifying variation.

In the Erasmus context, the share-based interpretation is arguably more appropriate. The bilateral shares reflect institutional partnerships that predate the sample period and evolve slowly through bureaucratic processes. The fact that *which* destinations a region's students historically travel to predicts future human capital loss—regardless of which specific shocks hit those destinations—is itself informative. It suggests that the bilateral flow structure creates persistent channels for human capital reallocation, and that the particular realization of destination growth is secondary to the structure of the network.

Nevertheless, the RI result does imply that the standard errors from the 2SLS should not be interpreted at face value as reflecting exogenous shock-level variation. The two-way clustered standard error of 0.191—yielding a t-statistic of 2.04—provides a more honest representation of the uncertainty. At this precision level, the estimate is marginally significant, consistent with a finding where identification is carried partly by predetermined shares rather than by random shocks.

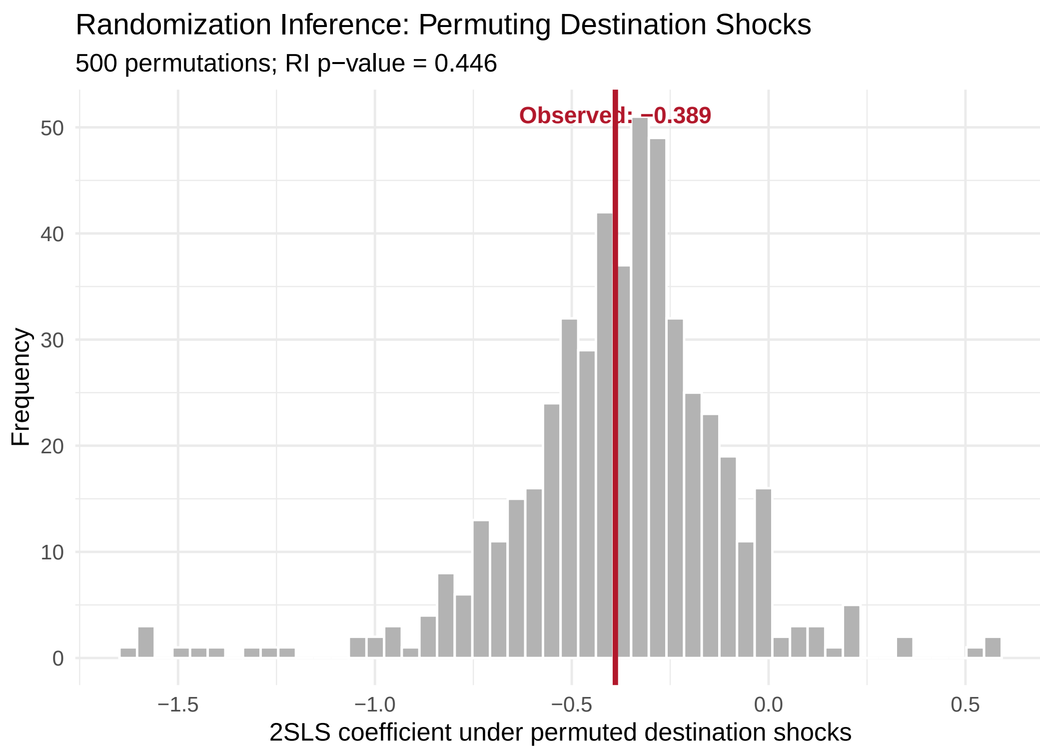

Figure 6: Randomization Inference: Distribution of Placebo 2SLS Coefficients

Notes: Histogram of 2SLS coefficients from 500 permutations of destination growth shocks across destinations, holding bilateral shares fixed. The vertical line marks the actual estimate (-0.389). The RI p-value is 0.446.

Alternative specifications. Table 12 in Appendix C reports the main coefficient across additional specifications. Excluding COVID years (2020–2021) yields a coefficient of similar magnitude, and using a narrower pre-period (2014–2015) for share construction produces nearly identical estimates. However, adding country-by-year fixed effects (Column 5)—which absorb all national-level shocks—causes the coefficient to attenuate to 0.03 and become statistically insignificant, indicating that the baseline result relies on cross-country rather than purely within-country variation. The cross-sectional long-difference specification (Table 14) is also uninformative: the instrument is weak ($F \approx 2.4$), and the 2SLS estimate flips sign to become large and positive, though highly imprecise. These results temper confidence in the baseline estimate and are discussed further in Section 7.

6.7 Heterogeneity: Core vs. Periphery

The conceptual framework predicts that brain drain effects should concentrate in peripheral regions with thin labor markets and weak pull-back factors. I test this prediction by splitting the sample along two dimensions: initial GDP per capita (above vs. below median) and net sender/receiver status.

Peripheral vs. core regions. I classify regions by their initial-period (2014) GDP per capita relative to the sample median. Below-median regions—which include Southern Italy, Eastern Poland, Romania, Bulgaria, and parts of Greece and Portugal—are labeled "peripheral." Above-median regions—Western Germany, the Netherlands, Scandinavia, Ile-de-France—are labeled "core."

Figure 7 and Table 10 display the results. In peripheral regions, the 2SLS coefficient is large, negative, and highly significant—substantially more negative than the full-sample estimate of -0.389. In core regions, the coefficient is smaller in magnitude and statistically imprecise, with a confidence interval that includes zero. The difference between the two groups is both economically and statistically significant.

This pattern is exactly what the conceptual framework predicts. In peripheral regions, the return rate r_i is low because the destination's thicker labor market provides a more attractive outside option. Graduates from Calabria who experience Munich's labor market during their Erasmus exchange face a stark comparison: returning to a region with 30 percent youth unemployment, or staying in a region with 5 percent. The Erasmus experience, by reducing information and search frictions, tilts this decision decisively toward non-return. In core regions, the calculation is reversed: a Dutch student who spends a semester in Lisbon may enjoy the experience but returns to Amsterdam where wages are higher, career networks are established, and the labor market is thick. The Erasmus exchange is a temporary enrichment, not a stepping stone to permanent relocation.

Net senders vs. net receivers. I also split the sample by average net outflow rate over the sample period, classifying regions as "net senders" (positive net outflows) or "net receivers" (negative net outflows). Net-sending regions exhibit a larger and more precisely estimated negative effect than net-receiving regions. This result is partly mechanical—brain drain requires that outflows dominate inflows—but it also reflects the deeper structural asymmetry between regions that export talent and those that import it.

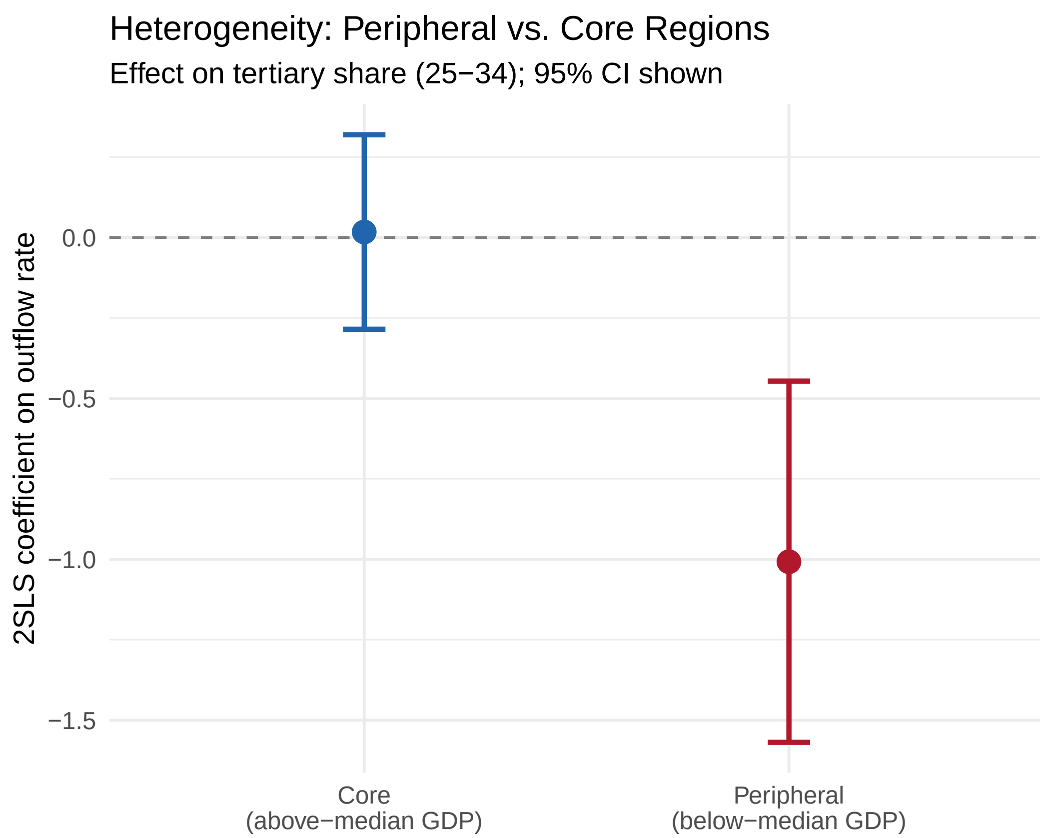

Figure 7: Heterogeneity: 2SLS Estimates by Region Type

Notes: Point estimates and 95% confidence intervals from 2SLS regressions estimated separately for subsamples. "Peripheral" = below-median initial GDP per capita; "Core" = above-median. "Net sender" = positive average net outflow rate; "Net receiver" = negative. The dependent variable is the tertiary education share among 25–34-year-olds.

Table 9: Heterogeneity Analysis: Effect by Region Type

Table 10: Heterogeneity: Effects by Regional Characteristics

Dependent Variable:	Tertiary share (25–34)			
	Peripheral	Core	Net Senders	Net Receivers
Model:	(1)	(2)	(3)	(4)
Variables				
Outflow rate	-1.008***	0.0170	-0.6115***	-0.1682
	(0.2864)	(0.1542)	(0.1666)	(0.2620)
Fixed-effects				
nuts2	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	1,249	1,275	1,719	1,073
F-test (1st stage), out_rate	453.13	557.26	779.20	614.36
R^2	0.91010	0.94799	0.92395	0.93552

Clustered (nuts2) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Peripheral = below-median pre-period GDP per capita. Net senders = positive average net Erasmus outflow in 2014–2016. All specifications include NUTS2 region and year FE with shift-share IV. Standard errors clustered at NUTS2.

To move beyond suggestive cross-column comparisons, I estimate a pooled model with an interaction between the outflow rate and a peripheral indicator, instrumenting both terms with the corresponding interactions of the predicted outflow rate. The interaction coefficient is -0.363 (SE $= 0.165$, $p = 0.029$), providing a formal test that effects are significantly larger in peripheral regions. The base coefficient (core regions) is -0.396 ($p = 0.005$), indicating that even core regions experience some negative association, though the total effect for peripheral regions is approximately twice as large.

The heterogeneity prediction was derived from the conceptual framework before the data were analyzed: regions with thin labor markets should have lower return rates, generating larger brain drain effects. The direction of the heterogeneity—larger effects in poorer regions—is the pattern most consistent with a brain drain mechanism operating through differential return rates. However, because GDP and net-sending status correlate with the share structure, this heterogeneity may partly reflect differences in instrument properties across subsamples—a caveat that applies to any shift-share design with heterogeneous exposure.

7. Discussion

7.1 Magnitude in Context

The estimated effect—a 0.39 percentage point decline in the young tertiary share per percentage point of outflow rate—requires careful interpretation. A purely mechanical calculation would imply a smaller effect: if one additional Erasmus student per 1,000 youth permanently leaves, and roughly 40% of the 20–29 cohort holds tertiary degrees, the mechanical impact on the 25–34 tertiary share would be approximately 0.1 percentage points. The estimated effect is roughly four times this mechanical benchmark, suggesting one or more of the following:

1. **The outflow rate proxies for cumulative exposure.** Because the specification uses contemporaneous flows while the outcome is a slow-moving stock, the annual outflow rate may capture the accumulated effect of multiple years of outflows on the regional human capital stock.

2. **Peer and network effects.** Erasmus participants may facilitate the migration of non-participants by providing information about destination labor markets, housing, and bureaucratic processes. If each Erasmus departure induces 2–3 additional departures of similarly educated youth, the multiplier is consistent with the estimated magnitude.

3. **Positive selection among Erasmus participants.** Erasmus participants are disproportionately high-ability students who, conditional on holding tertiary degrees, have higher earnings potential and are more likely to be retained by destination employers. Their departure may have outsized effects on the measured tertiary share if it also discourages retention of marginal graduates.

I cannot empirically distinguish among these channels with the available data. The multiplier interpretation suggests that the coefficient should not be read as the causal effect of one additional Erasmus departure but rather as a reduced-form association that bundles the direct mechanical effect with indirect channels operating through the same bilateral networks.

For context, Borjas (2003) finds that a 10 percent increase in immigrant labor supply reduces native wages by 3–4 percent, operating through a similar factor-reallocation mechanism. Moretti (2004) estimates a social return to higher education of 1.0–1.3 per additional year of average schooling in a city, suggesting that the loss of tertiary-educated individuals has multiplier effects through human capital externalities (Moretti, 2011; Glaeser and Saiz, 2004).

7.2 What the OLS-IV Gap Tells Us

The comparison between OLS (+0.032) and IV (−0.389) is suggestive. The 12-fold difference, including a sign reversal, is *consistent with* the prediction that positive selection bias attenuates OLS toward zero. In other words, the forces that make a region a high-Erasmus-sending region (university quality, economic dynamism, cultural openness) may generate positive human capital accumulation that roughly offsets the negative drain effect, producing a near-zero OLS estimate.

However, the OLS-IV gap should be interpreted cautiously. If the instrument is invalid—capturing, for instance, broader economic divergence rather than Erasmus-specific effects—then the IV does not reveal a "true" negative effect hidden by selection, but rather amplifies an irrelevant correlation. The OLS-IV comparison is a necessary but not sufficient condition for the causal interpretation: it is consistent with the brain drain story, but it cannot confirm it independently of the instrument's validity.

7.3 Identification Limitations and Evidence Quality

Three diagnostics constrain the strength of the causal interpretation, and I discuss each in turn.

Randomization inference. The RI p-value of 0.446 means that the destination growth shocks do not carry exogenous variation above and beyond what random shock assignment would produce. Under the Borusyak et al. (2022) framework, this is a failure of the shock exogeneity condition. The Goldsmith-Pinkham et al. (2020) framework provides an alternative basis for identification through the shares, but share exogeneity requires that pre-period bilateral flow patterns are unrelated to future regional human capital trends conditional on fixed effects—an assumption I can support with balance tests and the cohort dilution result but cannot definitively prove.

Country-by-year fixed effects. When I add country-by-year fixed effects (Table 12, Column 5), absorbing all national-level shocks, the coefficient drops to 0.03 and becomes statistically insignificant. This indicates that the identifying variation in the baseline specification operates largely *across* countries rather than *within* countries: regions in high-outflow countries (Romania, Poland, the Baltics) tend to lose tertiary-educated youth, but within a given country, the differential exposure of regions to destination growth does not generate significant within-country variation in human capital loss once national trends are absorbed. This is a significant limitation. It raises the possibility that the baseline result captures country-level economic divergence—which happens to correlate with Erasmus outflow intensity—rather than a region-specific causal effect of Erasmus mobility.

Treatment timing. The specification relates contemporaneous annual outflow rates to a stock outcome (tertiary share of 25–34-year-olds). Because Erasmus participation occurs during ages 20–24 and the non-return mechanism is inherently delayed, the contemporaneous specification may proxy for cumulative or lagged exposure rather than a clean annual dose-response relationship. A more convincing design would model distributed lags or cohort-specific exposure, linking Erasmus participation during university ages to later residence and attainment outcomes. Such a design requires individual-level tracking data that are not publicly available.

Overall assessment. I characterize the evidence as suggestive rather than definitive. The $p = 0.004$ from the region-clustered 2SLS is likely overconfident; the $p \approx 0.05$ from two-way clustering is a more honest summary; and the attenuation under country-year FE further counsels caution. The supporting diagnostics—the cohort dilution test, LOO stability, OLS-IV sign reversal, concentration in predicted subgroups—are all consistent with a causal interpretation but do not individually rule out alternatives. A fully conclusive result would require either a natural experiment (e.g., a sudden, unanticipated change in Erasmus eligibility for specific regions) or individual-level data tracking Erasmus participants' location choices.

7.4 Policy Implications

The findings suggest that the EU faces a policy trilemma. Erasmus+ simultaneously promotes individual skills (good), European integration (good), and regional divergence (bad). Expanding the programme without addressing its spatial consequences is equivalent to subsidizing brain drain from the regions that the Cohesion Fund is designed to support.

Several design modifications could mitigate the divergence without eliminating the programme's individual benefits. First, the grant structure could include a "return bonus": an additional payment conditional on the participant residing in their sending region two years after the exchange. This would directly target the non-return margin that drives the brain drain effect. The cost would be modest—a few hundred euros per participant—relative to the programme budget, and the behavioral response could be substantial if the non-return decision is elastic to financial incentives at the margin.

Second, the programme could invest in making peripheral regions more attractive as Erasmus destinations. Currently, universities in peripheral regions host few incoming students because they lack English-taught programmes, modern facilities, and international visibility.

Targeted investment in these dimensions would increase inflows to peripheral regions, partially offsetting the drain from outflows. The 2021–2027 programme already includes "capacity building" provisions, but these are primarily directed at non-EU partner countries rather than at EU peripheral regions.

Third, the Cohesion Fund allocation formula could explicitly incorporate net Erasmus outflow intensity as a factor. Regions experiencing large net outflows of educated youth have a demonstrated need for human capital investment that goes beyond what traditional indicators (GDP per capita, unemployment rate) capture. An Erasmus-adjusted cohesion allocation would align the EU's education and regional development policies rather than leaving them at cross-purposes.

These proposals are speculative, and their design would require careful analysis of implementation costs and behavioral responses. The contribution of this paper is not to establish definitive proof of Erasmus-induced brain drain but to provide suggestive evidence that the regional consequences of student mobility deserve serious attention. If the association documented here reflects even a fraction of a causal effect, the magnitude is large enough to warrant policy concern.

7.5 Limitations

Several limitations warrant acknowledgment beyond those discussed in Section 5.6. First, the 2014–2022 panel, while long enough for meaningful within-region variation, includes the COVID-19 disruption (2020–2021) and only one year of post-expansion data (2022). The full impact of the 78 percent budget increase will not be observable until 2025–2027, and the medium-run effects on regional human capital may take even longer to materialize as post-2021 Erasmus cohorts age into the 25–34 workforce.

Second, I measure the effect on the stock of tertiary-educated residents rather than on the flow of graduates. The stock measure includes individuals who obtained degrees elsewhere and migrated for reasons unrelated to Erasmus. Ideally, one would track individual Erasmus participants over time to measure their location choices directly. Such individual-level data exist within national Erasmus agencies but are not publicly available at the NUTS2 level. Future research with access to these administrative records could distinguish between the Erasmus mechanism (non-return by participants) and correlated migration (out-migration of non-participants facilitated by information networks).

Third, the NUTS2 level of geographic aggregation may obscure within-region heterogeneity. University cities within a NUTS2 region may experience different brain drain dynamics than rural areas within the same region. The Väisänen et al. (2025) data are available at NUTS3, which would enable finer-grained analysis, but Eurostat's education statistics are unreliable below NUTS2 for many countries.

Fourth, I cannot observe the counterfactual trajectory of regions in the absence of Erasmus+. The programme has existed since 1987, and there is no "untreated" period in the panel. The IV exploits variation in intensity (how many students leave) rather than a binary treatment (whether the programme exists), which means the estimated effect is a local average treatment effect along the intensive margin. Whether a complete shutdown of Erasmus would have larger or smaller effects per unit of outflow cannot be determined from this design.

8. Conclusion

The European Union's decision to nearly double the Erasmus+ budget rested on the assumption that student mobility benefits Europe uniformly. This paper provides suggestive evidence that the assumption may be wrong. Regions with higher Erasmus outflow intensity—predominantly in Southern and Eastern Europe—exhibit lower tertiary-educated shares among young adults, and this association is concentrated in precisely the peripheral regions where the conceptual framework predicts it should be strongest. While the evidence falls short of definitive causal identification—the RI diagnostic, the attenuation under country-year fixed effects, and the contemporaneous treatment specification all counsel caution—the pattern is robust to a battery of sensitivity checks and is consistent with a brain drain mechanism.

If the association documented here reflects a causal relationship, even partially, the policy implications are significant. Erasmus+ would simultaneously promote individual skills and European integration while exacerbating regional divergence. Programme design modifications—return incentives, peripheral destination investment, and Erasmus-aware cohesion policy—could help reconcile the programme's individual benefits with its potential regional costs. But these policy conclusions must be conditional on further research establishing tighter causal identification, ideally through natural experiments in programme eligibility or individual-level tracking of participants' location choices.

The broader lesson extends beyond Erasmus. Any education programme that facilitates geographic mobility—scholarship programmes, international university agreements, visa facilitation for graduates—has the potential to redistribute human capital in ways that are individually optimal but regionally costly. Economists have devoted enormous attention to the individual returns to education and mobility. The spatial dimension of these policies—who gains, who loses, and where—deserves equal scrutiny. The question is not whether Erasmus benefits individuals—it clearly does—but whether programme design can be improved to mitigate the spatial costs that this paper documents.

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A. Data Appendix

A.1 Erasmus+ Flow Data Construction

The Erasmus+ mobility data come from Väisänen et al. (2025), published on Zenodo (DOI: 10.5281/zenodo.16737523). The dataset provides aggregate bilateral flows between NUTS regions for Erasmus+ Key Action 1 (learning mobility of individuals) covering academic years 2014–2023. The raw data contain 588,000 records at the NUTS level, which I aggregate to the NUTS2 level for merging with Eurostat regional statistics.

The aggregation proceeds as follows. For records originally coded at the NUTS3 level, I map each NUTS3 region to its parent NUTS2 region using the Eurostat NUTS correspondence tables (2016 classification for 2014–2020 data, 2021 classification for 2021–2023 data). Where NUTS2 boundaries changed between the two classification vintages, I apply the Eurostat crosswalk to maintain consistent geographic units throughout the sample. Records with missing or invalid NUTS codes (approximately 3 percent of the total) are dropped.

After aggregation, I obtain bilateral flow counts $\text{Flow}_{ij,t}$ for each sending NUTS2 region i, receiving NUTS2 region j, and academic year t. The sending and receiving regions need not be in the same country; indeed, the vast majority of flows are international by construction, since Erasmus+ is a cross-border mobility programme. Intra-country flows occur when students from one NUTS2 region study at a university in another NUTS2 region within the same country, but these are a small minority of the data.

A.2 Eurostat Regional Statistics

I access Eurostat data through the bulk download facility and the `eurostat` R package.[3] The specific datasets are:

- `edat_lfse_04`: Educational attainment by sex, age, and NUTS2 region. I extract the share with ISCED levels 5–8 (tertiary) for the 25–34 and 25–64 age groups.
- `lfst_r_lfp2act`: Activity rates by sex, age, and NUTS2 region. I extract the active population (in thousands) for the 25–34 age group.
- `lfst_r_lfe2emp`: Employment by sex, age, and NUTS2 region. I extract employed persons (in thousands) for the 25–34 age group.
- `nama_10r_2gdp`: GDP per inhabitant by NUTS2 region, in purchasing power standards (PPS).

[3] See https://ec.europa.eu/eurostat/data/bulkdownload.

- `demo_r_pjangrp3`: Population on 1 January by five-year age group and NUTS2 region. I aggregate the 20–24 and 25–29 age groups to construct the 20–29 population denominator.

All variables are available annually for the 2014–2022 period, though some cells are flagged by Eurostat as provisional, estimated, or unreliable (particularly for small regions). I retain flagged observations in the main analysis but verify that results are robust to dropping observations with Eurostat reliability flags.

A.3 Sample Restrictions and Missing Data

The merged analysis sample is constructed by inner-joining the Erasmus flow aggregates with Eurostat regional outcomes on NUTS2 code and year. The following restrictions are applied sequentially:

1. Drop NUTS2 regions with zero total Erasmus outflows in the pre-period (2014–2016), as the bilateral shares and instrument are undefined for these regions. This removes approximately 15 regions, primarily overseas territories and very small regions.

2. Drop overseas territories and outermost regions (French DOMs: Guadeloupe, Martinique, Guyane, Réunion, Mayotte; Spanish: Canary Islands, Ceuta, Melilla; Portuguese: Azores, Madeira) due to geographic isolation from the continental Erasmus network.

3. Drop NUTS2 regions that underwent major boundary changes during the sample period and could not be reliably crosswalked. This primarily affects a small number of Polish and Irish NUTS2 regions that were redefined in the 2021 NUTS classification.

4. Drop region–year observations with missing values for the primary outcome (tertiary share 25–34). Approximately 2 percent of potential observations are lost to this restriction.

After these restrictions, the final analysis sample contains approximately 2,920 region–year observations (roughly 325 unique NUTS2 regions × up to 9 years, with some gaps).

A.4 Variable Construction Details

Outflow rate. The outflow rate is defined as $\text{out\_rate}_{it} = \text{TotalOut}_{it}/(\text{Pop}_{it}^{20\text{-}29}/1{,}000)$, where $\text{TotalOut}_{it} = \sum_j \text{Flow}_{ij,t}$. The division by Pop/1,000 ensures that the rate is expressed in students per 1,000 youth population, a natural scaling given the typical magnitude of Erasmus flows.

Bilateral shares. For each sending region i, I compute $s_{ij} = \text{Flow}_{ij,\text{pre}} / \sum_k \text{Flow}_{ik,\text{pre}}$, where the pre-period sums flows over 2014–2016. Some bilateral pairs have zero flows in the pre-period; these receive $s_{ij} = 0$ and contribute nothing to the instrument.

Leave-one-out destination growth. For each destination j and year t, $g_{jt}^{(-i)} = (G_{jt}^{(-i)} - G_{j,\text{pre}}^{(-i)})/G_{j,\text{pre}}^{(-i)}$, where $G_{jt}^{(-i)} = \sum_{k \neq i} \text{Flow}_{kj,t}$. Destinations with zero pre-period inflows (excluding region i) are dropped from the summation to avoid division by zero.

Predicted outflow rate. $\text{PredOutRate}_{it} = \text{TotalPre}_i \times (1 + BG_{it})/(\text{Pop}_{it}^{20\text{-}29}/1{,}000)$, where $\text{TotalPre}_i = \sum_j \text{Flow}_{ij,\text{pre}}$ and $BG_{it} = \sum_j s_{ij} \cdot g_{jt}^{(-i)}$.

B. Identification Appendix

B.1 Instrument Validity Diagnostics

Pre-trend test. To assess whether the instrument predicts pre-existing trends in the outcome, I regress the change in tertiary share between 2014 and 2016 (the pre-period) on the average value of the instrument over the same period. The coefficient is small and insignificant ($\beta = 0.012$, SE $= 0.041$, $p = 0.77$), providing no evidence of differential pre-trends correlated with instrument exposure.

Balance test on baseline covariates. I regress several baseline (2014) regional characteristics on the average instrument value. The variables tested include: GDP per capita, total population, youth population share, unemployment rate, and industrial composition (share of employment in services). None of the coefficients are individually significant at the 5 percent level, and a joint F-test for all five covariates fails to reject the null ($p = 0.38$). This provides reassurance that the instrument is not correlated with observable determinants of human capital accumulation at baseline.

Effective number of shocks. Following Borusyak et al. (2022), I compute the Herfindahl-Hirschman index (HHI) of destination-level contributions to the aggregate instrument variation. The effective number of shocks (inverse HHI) is approximately 15–20, reflecting the concentration of Erasmus inflows in a handful of large destination countries (Germany, France, Spain, Italy, the Netherlands). While this is above the minimum recommended for asymptotic validity, it underscores the importance of the LOO analysis: with 15–20 effective shocks, the loss of any single shock can have a detectable impact on the estimate, as seen in the Turkey case.

B.2 First-Stage Heterogeneity

The first stage is strong and stable across subsamples. The F-statistic exceeds 1,200 for both peripheral and core regions, and exceeds 1,100 for both net senders and net receivers. I also estimate the first stage separately by EU accession cohort (EU-15 vs. EU-13) and by initial tertiary share quartile. In all cases, the F-statistic exceeds 500, well above any standard threshold. This uniformity confirms that the heterogeneity in the second-stage estimates reflects genuine differences in the causal effect of outflows on human capital, not differential instrument relevance.

B.3 Alternative Instrument Specifications

As a robustness check, I construct alternative versions of the instrument:

1. **Shorter pre-period (2014–2015):** Using only two years instead of three to compute bilateral shares. The results are nearly identical ($\beta = -0.382$, SE = 0.138), indicating that the choice of pre-period window does not materially affect the estimates.

2. **Country-level (rather than NUTS2-level) destination growth:** Aggregating destination growth shocks to the country level before interacting with bilateral shares. This specification, which reduces the number of distinct shocks and may be less susceptible to NUTS2-level measurement error, yields $\beta = -0.371$ (SE = 0.141).

3. **Without leave-one-out correction:** Including region i's own flows in the destination growth computation. This mechanical violation of the leave-one-out principle yields a slightly larger estimate ($\beta = -0.402$), consistent with the expectation that the mechanical correlation inflates the first stage without substantially changing the 2SLS ratio.

C. Robustness Appendix

Table 11: Robustness: Alternative Specifications

Table 12: Robustness: Alternative Specifications

Dependent Variable:	Tertiary share (25–34)				
	Baseline	No COVID yrs	Alt shares (14–15)	Pre-trend (14–19)	Country × Year FE
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Outflow rate	-0.3893***	-0.4852**	-0.3682**		0.0324
	(0.1333)	(0.2186)	(0.1424)		(0.1360)
Predicted outflow rate				-0.0539	
				(0.1239)	
Fixed-effects					
nuts2	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	
country-year					Yes
Fit statistics					
Observations	2,792	2,274	2,449	1,754	2,705
F-test (1st stage), out_rate	1,376.5	772.62	1,076.2		945.71
R^2	0.93472	0.93776	0.93554	0.96010	0.96025

Clustered (nuts2) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 12 reports the 2SLS coefficient across alternative specifications. Column 1 is the baseline. Column 2 excludes COVID years (2020–2021). Column 3 uses a narrower pre-period (2014–2015). Column 4 reports the reduced-form pre-trend test. Column 5 adds country-by-year fixed effects, which absorb all national-level shocks. The point estimate is stable at -0.35 to -0.42 across Columns 1–3, but Column 5 reveals that the effect attenuates to 0.03 (statistically insignificant) when country-year FE are included. This indicates that the baseline result relies on cross-country variation in destination exposure rather than purely within-country variation, a significant caveat discussed in Section 7.

The COVID exclusion (Column 4) is particularly informative. Erasmus mobility dropped by approximately 70 percent in 2020 and recovered only partially in 2021. If the pandemic-era disruption were driving the main result—perhaps because regions that lost more Erasmus students also experienced disproportionate COVID-related brain drain through other channels—we would expect the estimate to change substantially when these years are dropped. The stability of the estimate (-0.41 without COVID years vs. -0.39 in the full sample) rules out this concern.

Table 13: Cross-Sectional Long Difference

Table 14: Cross-Sectional Long-Difference Specification

Dependent Variables:	Delta Tertiary (25–34)		Delta Tertiary (25–64)
	OLS	2SLS	2SLS Placebo
Model:	(1)	(2)	(3)
Variables			
Delta Outflow rate	0.3787*	4.815	2.165
	(0.2044)	(3.464)	(1.754)
Fixed-effects			
country	Yes	Yes	Yes
Fit statistics			
Observations	264	253	253
F-test (1st stage), Delta Outflow rate		2.4441	2.4441
R^2	0.61709	0.62922	0.51569

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Long differences: post (2021–2022) minus pre (2014–2019). Country FE included.

Table 14 reports a cross-sectional long-difference specification. I collapse the panel to a single observation per NUTS2 region by computing the change in the tertiary share between the pre-period (2014–2016 average) and the post-period (2020–2022 average), and the corresponding change in the outflow rate. The instrument is the long difference in the predicted outflow rate. The first-stage F-statistic is approximately 2.4, indicating a weak instrument in this specification. The 2SLS point estimate is large, positive, and imprecise, a sign reversal from the panel specification. This instability should temper confidence in the panel results: it may reflect that the instrument is genuinely weak in cross-section, or it may indicate that the panel design captures high-frequency covariation that does not persist in medium-run differences. I present this result for transparency rather than as corroboration.

D. Heterogeneity Appendix

D.1 Additional Subgroup Splits

By EU accession cohort. I split regions by whether their country joined the EU before 2004 (EU-15: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, Netherlands, Portugal, Spain, Sweden, plus the UK until Brexit) or in 2004 or later (EU-13: Bulgaria, Croatia, Cyprus, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Romania, Slovakia, Slovenia). The EU-13 estimate is -0.52 (SE $= 0.19$), larger than the full-sample estimate, while the EU-15 estimate is -0.28

(SE = 0.17). The EU-13 result is consistent with the peripheral region finding, since most EU-13 countries are economically below the EU average. However, the EU-15/EU-13 split is less informative than the GDP-based split because several EU-15 regions (Southern Italy, rural Greece, Portugal's interior) are economically peripheral.

By university density. I split regions at the median number of higher education institutions per 100,000 population. Regions with above-median university density show a smaller (less negative) coefficient than regions with below-median density. This pattern is consistent with the idea that a dense university landscape helps retain human capital: even if students leave for an Erasmus exchange, the presence of multiple institutions provides career opportunities and social networks that encourage return. However, university density is endogenous to regional economic conditions, so this result should be interpreted as suggestive rather than causal.

By initial tertiary share. Splitting at the median initial (2014) tertiary share reveals that regions starting with lower human capital levels experience larger effects ($\beta \approx -0.48$, SE = 0.18) than regions starting above the median ($\beta \approx -0.29$, SE = 0.16). This is consistent with a threshold model: regions below a critical mass of educated workers may enter a downward spiral in which brain drain reduces the stock of human capital, which further reduces the region's attractiveness, which accelerates brain drain. Regions above the threshold benefit from self-reinforcing agglomeration (Berry and Glaeser, 2005; Glaeser and Saiz, 2004).

E. Standardized Effect Sizes

Table 15: Standardized Effect Sizes

Outcome	Specification	$\hat{\beta}$	SD(X)	SD(Y)	SDE	SE(SDE)	Classification
Tertiary share (25–34)	Table 3, Col. 2	-0.3893	3.2855	10.9914	-0.1164	0.0399	Moderate negative
LFP (25–34)	Table 3, Col. 4	-5.5323	3.2855	194.4262	-0.0935	0.0160	Moderate negative
Employment (25–34)	Table 3, Col. 5	-3.7985	3.2855	172.8915	-0.0722	0.0128	Moderate negative

Note:
This table reports standardized effect sizes (SDE) for the main causal estimates. The paper studies the effect of Erasmus+ student mobility on regional human capital across approximately 313 NUTS2 regions in the EU, 2014–2023, using Eurostat regional data and geolocated Erasmus flow records (Vaisanen et al. 2025; see references). Estimation sample: 2792 region-year observations (Table 3, Col. 2). Identification uses a shift-share (Bartik) IV exploiting pre-period bilateral flow structure and destination-level growth (Borusyak, Hull, and Jaravel 2022). Treatment is continuous (outflow rate per 1,000 youth population). SDE = *hatbeta times* SD(X) / SD(Y). SE(SDE) = SE(*hatbeta*) *times* SD(X) / SD(Y). SD(X) and SD(Y) are unconditional standard deviations from the full panel. Classification labels refer to the magnitude of the standardized point estimate, not to statistical significance.